

Multi-Task Deep Learning Framework for Unified Cattle Health and Behavior Monitoring

by

Hasin Ishrak

22201133

Namira Abrar Haque

22201191

Sanjida Akter Bithi

22201180

Shouvik Banik

23101295

Nusrat Lamya Faruk

24241182

A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University
September 2026

© 2026. Brac University
All rights reserved.

Declaration

It is hereby declared that

1. The thesis submitted is my/our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

Student's Full Name & Signature:

Hasin Ishrak
22201133

Namira Abrar Haque
22201191

Sanjida Akter Bithi
22201180

Shouvik Banik
23101295

Nusrat Lamya Faruk
24241182

Approval

The thesis titled “Multi-Task Deep Learning Framework for Unified Cattle Health and Behavior Monitoring” submitted by

1. Hasin Ishrak (22201133)
2. Namira Abrar Haque (22201191)
3. Sanjida Akter Bithi (22201180)
4. Shouvik Banik (23101295)
5. Nusrat Lamya Faruk (24241182)

of Fall, 2026 has been accepted as satisfactory in partial fulfillment of the requirement for the degree of B.Sc. in Computer Science and Engineering on September, 2026.

Examining Committee:

Supervisor:
(Member)

Dr. Md. Khalilur Rahman
Professor
Department of Computer Science and
Engineering
Brac University

Co-Supervisor 1:
(Member)

Mehedi Hasan Emo
Lecturer
Department of Computer Science and
Engineering
Brac University

Co-Supervisor 2:
(Member)

Mollah MD Saif
Lecturer
Department of Computer Science and
Engineering
Brac University

Thesis Coordinator:
(Member)

Dr. Md. Golam Rabiul Alam
Professor
Department of Computer Science and
Engineering
Brac University

Head of Department:
(Chair)

Dr. Sadia Hamid Kazi
Associate Professor
Department of Computer Science and
Engineering
Brac University

Ethics Statement

This research is one of the requirements of the B.Sc. degree in Computer Science and Engineering. The sources of all data used in the research, as well as benchmarks used in computations, have been properly cited in the bibliography.

The research was based only on public data available on the internet that contains pictures and video of cows. The data includes ScienceDB, CVB, Kaggle Beef, and SideViewCows2026 datasets. Invasive experiments on live animals have not been conducted. The aim of the research is to validate automated vision techniques and multitask deep learning models. The outcome of the experiment cannot be regarded as evidence of clinical efficiency or justification of autonomous therapy.

The individuals and institutions that contributed to this research in terms of support and assistance have received proper recognition. The research followed the university's ethical code of conduct and posed no ethical issues.

Abstract

Continuous precision cattle health monitoring involves assessing BCS, behavior recognition, and re-ID of cattle. These tasks involve different kinds of visual information such as body morphology, temporal motion, and fine-grained biometric appearance. This work aims to examine cattle-centric visual representation through target localization, segmentation masks, and lightweight temporal modeling under leakage-aware single-task and multi-task settings.

Single-task experiments showed task-specific improvement. The cattle-oriented model provided improved performance of BCS and behavior recognition, while open set Re-ID performance increased with the oracle mask setting. Hard parameter sharing using multi-task learning degraded the results in all the three tasks. Nonetheless, the private residual adapters provided selective improvements but did not consistently improve the degraded results. PCGrad training showed frequently conflicting gradients of the shared backbone. Gradient projection provided a way to recover selectively from the conflicting gradients especially for behavior recognition. These findings show that shared representations do not automatically benefit heterogeneous cattle-monitoring tasks. Effective unified monitoring therefore requires balancing shared cattle-centered features with task-specific representation capacity and conflict-aware optimization.

Keywords: cattle monitoring; body condition scoring; behavior recognition; cattle re-identification; cattle-centered representation; multi-task learning; negative transfer.

Dedication

Dedicated to our beloved parents and families, whose unconditional love, sacrifices, endless patience, and continuous prayers have been our greatest source of strength throughout this academic journey.

Acknowledgement

First of all, all praises to Almighty Allah for giving us the power, determination, and endurance to finish this undergraduate thesis.

We would like to take this opportunity to thank our supervisor, **Dr. Md. Khalilur Rahman**, Professor, Department of Computer Science and Engineering, Brac University, for all the valuable help that he has provided during this research project. His knowledge about computer vision, robotics and machine intelligence was of great assistance to us in finishing the work.

We would also like to sincerely thank our co-supervisors, **Mehedi Hasan Emo**, Lecturer, and **Mollah MD Saif**, Lecturer, Department of Computer Science and Engineering, Brac University, for their valuable feedback and support during the evaluation of our research.

It is our great pleasure to thank the Department of Computer Science and Engineering, Brac University, for the provision of state-of-the-art computational facilities and an enabling academic atmosphere for our research. We also take this opportunity to extend our heartfelt gratitude to the authors and institutions behind the benchmark datasets used in this research, without whose open access efforts this study would not have been possible.

Lastly, our deepest appreciation goes out to our parents, families, and friends who have provided us with constant love, guidance, and prayers through our entire academic experience.

Table of Contents

Declaration	i
Approval	ii
Ethics Statement	iv
Abstract	v
Dedication	vi
Acknowledgment	vii
Table of Contents	viii
List of Figures	xi
List of Tables	xii
Nomenclature	xiii
1 Introduction	1
1.1 Background	1
1.2 Rationale of the Study and Motivation	2
1.2.1 Task-Appropriate Visual Representations	2
1.2.2 Preliminary Investigations and Their Limitations	3
1.2.3 Rationale for Jointly Studying the Three Tasks	3
1.3 Problem Statement	3
1.4 Objectives	4
1.5 Methodology in Brief	5
1.6 Scopes and Challenges	6
2 Literature Review	7
2.1 Preliminaries	7
2.1.1 Visual Representations and Transfer Learning	7
2.1.2 Prediction Tasks and Their Visual Requirements	8
2.1.3 Multi-Task Learning and Negative Transfer	8
2.2 Review of Existing Research	9
2.2.1 From Automated Observations to Integrated Livestock Monitoring	9

2.2.2	Body Condition Scoring: Anatomy, Image Geometry, and Ordered Prediction	9
2.2.3	Behavior Recognition: Posture, Motion, and Environmental Context	10
2.2.4	Individual Identification and Re-ID: From Coat Patterns to Cross-Setting Recognition	11
2.2.5	Localization and Segmentation as Representation Design . . .	12
2.2.6	Pose, Anatomy, and Viewpoint	14
2.2.7	Temporal Representations for Activity Recognition	15
2.2.8	Shortcut Learning, Domain Shift, and Evaluation	16
2.2.9	Multi-Task Learning: Selective Sharing and Negative Transfer	17
2.3	Summary of Key Findings	18
2.3.1	A Common Animal, but Different Information Requirements .	19
2.3.2	Representation Quality Must Be Connected to Robust Evaluation	19
2.3.3	Selective Sharing as the Basis of the Unified Framework	19
3	Requirements, Impacts and Constraints	21
3.1	Final Specifications and Requirements	21
3.2	Societal Impact	22
3.3	Environmental Impact	23
3.4	Ethical Issues	23
3.5	Standards and Technical Conventions	23
3.6	Project Management Plan	24
3.7	Risk Management	25
3.8	Economic Analysis	26
4	Proposed Methodology	27
4.1	Design Process and Methodology Overview	27
4.2	Preliminary Designs and Model Specifications	29
4.2.1	Representation Alternatives	29
4.2.2	Baseline RGB Body Condition Scoring	29
4.2.3	Perception-Enhanced Body Condition Scoring	30
4.2.4	Baseline Single-Frame RGB Behavior Recognition	30
4.2.5	Perception-Enhanced Temporal Behavior Recognition	31
4.2.6	Baseline RGB Cattle Re-Identification	31
4.2.7	Oracle Segmentation-Guided Re-Identification	32
4.3	Data Collection and Preparation	32
4.3.1	Data Sources and Scientific Roles	32
4.3.2	Label Harmonization and Split Protection	34
4.3.3	Perception Feasibility and Component Selection	34
4.4	Implementation of Selected Design	35
4.4.1	Monolithic Hard-Shared Multi-Task Control Architecture . . .	35
4.4.2	Modular Task-Private Multi-Task Architecture	36
4.4.3	Gradient-Projected Hard-Shared Multi-Task Optimization . .	38
4.4.4	Multi-Task Joint Optimization Protocol	40
4.4.5	Architectural Gradient Isolation and Claim Boundaries	43
4.4.6	Evaluation Metrics and Protocol Harmonization	43

5	Result Analysis	46
5.1	Performance Evaluation	46
5.1.1	Body Condition Scoring Results	46
5.1.2	Behavior Recognition Results	47
5.1.3	Cattle Re-Identification Results	48
5.1.4	Perception Coverage and Supporting Viewpoint Evidence . . .	49
5.2	Multi-Task Learning Results	50
5.2.1	Body Condition Scoring Multi-Task Outcomes	50
5.2.2	Behavior Recognition Multi-Task Outcomes	51
5.2.3	Cattle Re-Identification Multi-Task Outcomes	52
5.2.4	Observed Shared-Backbone Gradient Conflict	53
5.3	Analysis of Multi-Task Design Solutions	53
5.4	Statistical Analysis and Uncertainty Boundaries	54
5.5	Cross-Task Relationships and Claim Boundaries	56
5.6	Discussion of Findings	56
6	Conclusion	58
6.1	Summary of Findings	58
6.2	Answers to the Research Questions	59
6.2.1	Research Question 1	60
6.2.2	Research Question 2	60
6.2.3	Research Question 3	61
6.3	Contributions of the Thesis	62
6.4	Limitations	63
6.5	Future Work	65
6.6	Conclusion	66
	Bibliography	67
A	Dataset and Experiment Evidence	72
A.1	Versioned Evidence Register	72
A.2	Canonical Evaluation Boundaries	74
A.3	Remaining Evidence Boundaries	74
A.4	Model Names and Repository Identifiers	74
A.5	Methodology Evidence Cross-Reference	75
B	Reproducibility and Evidence Boundaries	76
B.1	Reproducibility Boundary	76
B.2	Scientific Interpretation Boundaries	76
B.3	Open Deliverables	77

List of Figures

4.1	Comparative research design	27
4.2	Task-specific cattle-centered input pipelines	29
4.3	Multi-task architecture comparison: hard-shared vs. modular	37
4.4	Multi-task training super-step protocol and gradient routing	44
5.1	Run 2 Behavior recall by class	47

List of Tables

2.1	Visual information and task requirements	15
2.2	Approaches to managing multi-task interaction	18
3.1	Research requirements	21
3.2	Research milestones	24
3.3	Research risks and mitigations	25
3.4	Recorded runtime and partial costs	26
4.1	Experimental model configurations	28
4.2	Dataset roles and interpretation boundaries	33
4.3	Canonical split and retrieval protocol sizes	34
4.4	Selected perception feasibility results	35
4.5	Architectural specification of multi-task configurations	41
4.6	Multi-task training protocol and optimization hyperparameters	42
5.1	BCS RGB and cattle-centered comparison	46
5.2	Behavior RGB and cattle-centered temporal comparison	48
5.3	Matched Behavior class-wise comparison	48
5.4	Matched Behavior source-specific comparison	48
5.5	Barn-to-Parlor Re-ID results	49
5.6	Snapshot-to-Parlor Re-ID results	49
5.7	Perception coverage in completed enhanced evaluations	50
5.8	Multi-task Body Condition Scoring results	51
5.9	Multi-task Behavior Recognition results	51
5.10	Multi-task Behavior source-disjoint breakdown	52
5.11	Multi-task Cattle Re-Identification results under SideView Protocol A	52
5.12	PCGrad shared-backbone gradient conflict diagnostics	53
5.13	Evidence-led design decisions across single-task and multi-task investigations	55

Nomenclature

The following abbreviations are used in this report.

BCE Binary Cross-Entropy

BCS Body Condition Scoring

CNN Convolutional Neural Network

DGE Depth-Grayscale-Edge representation

GT Ground truth

IoU Intersection over Union

MAE Mean Absolute Error

MTL Multi-Task Learning

Re-ID Re-Identification

RGB Red, Green and Blue image channels

TCN Temporal Convolutional Network

Chapter 1

Introduction

1.1 Background

The monitoring of cattle involves knowledge concerning the state and behavior of the animal and any variations with time. The cow may reduce its intake, have peculiar resting behavior, or become gradually thin. It is necessary to interpret such observations by associating them with the particular cow. This explains why animal-based measures must be evaluated along with housing and management parameters in animal welfare research [3]. BCS, Behavior Detection, and individual identification are some of the measures that provide relevant information for this objective. BCS is utilized for determining the body reserves of a cow through visual observation of the amount of body tissues covering critical parts of the body. Unlike physical measurements of the body, BCS focuses on the inspection of selected body regions such as tailhead, spine, loin, and pelvis [1], [2].

Behavior detection involves behaviors such as eating, drinking, standing, lying down, and walking. Animal identification allows repeated observations to be linked with the same animal. Both tasks together provide a better understanding of the animal compared to using either of them alone. The process of gathering information on the animal is a very time-consuming activity that even requires some level of expertise. Short-term observation may lead to missing important welfare events. In precision livestock farming, sensors and data analysis help to observe the animals more frequently and continuously. According to Sani et al. [48] and Lee et al. [47], it is not an easy task to make the collected data useful for farm management decisions. For that, the observations need to be reliable and linked to the correct animal over time.

Computer vision is an effective method of monitoring the appearance, posture, and movements of the animal without attaching sensors to each animal individually. Review articles on vision-based livestock monitoring have shown multiple examples of computer vision application on dairy farms and in other livestock systems. However, they also highlight difficulties in applying the models when conditions are not the same as those during training [38], [51]. Therefore, the importance of computer vision is not limited to automating one specific task but rather includes its capability to obtain multiple measurements from one picture or video frame. With deep learning, visual features can be extracted directly from data while reusable image encoder models such as ResNet and EfficientNet lessen the need to design a new feature extractor for each task [4], [18]. Earlier researches on cows have

used learned visual features for body condition scoring, behavior recognition, and individual animal identification.

Combining such tasks is complicated by the fact that each of them could require different types of visual information. Multi-task learning (MTL) consists of training several tasks simultaneously, sharing some components of the neural network while maintaining separate outputs for each task. Shared components might be helpful since tasks would be able to reuse useful features, but at the same time, they could lower performance when the shared features are not appropriate for some tasks. Therefore, research related to task grouping and selective sharing recommends testing relationships between tasks rather than assuming them [5], [22]. This thesis investigates this problem for cattle body condition, behavior, and identity.

1.2 Rationale of the Study and Motivation

1.2.1 Task-Appropriate Visual Representations

A typical MTL architecture consists of a single image encoder and multiple prediction heads. This design is simple and efficient. However, even if tasks use the same image, they could still require different types of visual information.

Two tasks might work on the same cow and yet require different kinds of visual cues. This research is driven by the need to handle such situations in one system. BCS mainly depends on the cow’s body form and visual cues about body condition. Behavior recognition requires posture and movement cues along with some contextual cues about the surroundings. The re-identification task requires distinguishing features which make one cow different from another, for instance, the coat pattern and body form. Thus, a visual cue that is useful for one task may distract another task or even mislead it. For instance, the coat pattern can be helpful in discriminating between individuals; however, it should not become a shortcut for BCS estimation. Likewise, the feeding place can be useful in recognizing an action; however, a re-identification algorithm should not identify the cow only because it is normally observed in a certain place.

This problem is associated with shortcut learning. Shortcut learning is when a model learns an easy correlation, rather than the relationship we actually want it to learn. This problem is described by Geirhos et al. [21] in deep learning and image backgrounds can strongly affect recognition as shown by Xiao et al. [27]. The cow and its surrounding environment are present in farm images, so it is important to understand what information the model uses, especially if the camera, farm or recording condition is changed.

In this thesis, learning from *cattle-centred representations* pertains to arranging the visual inputs according to the animal and requirements of each task. Localization serves as a means of finding the target cow, while segmentation isolates the animal from the background. The body structure may be encoded by means of pose or anatomical information, and video allows one to track postural and movement changes. Such cues serve as possibilities for experiments and not as features that always help in performing better in all tasks. The primary goal is the search for the shared information, the specific task information, and whether context may also be helpful.

1.2.2 Preliminary Investigations and Their Limitations

During the first stages of research, a typical pretrained visual backbone with several prediction heads was applied. This multitasking system performed poorly as compared to a single tasking system when applied in body condition and behavior situations.

The conclusion drawn is that just adding outputs to a shared network is insufficient. This is why we decided to consider the information required at each step in more detail. The analysis of the data showed that some images may not be independent of each other and may correspond to the same cow passage, neighboring frames from one video, or the same event shot from different angles simultaneously. A passage label or temporary tracking ID does not necessarily correspond to the identity of the cow. All this impacts the effectiveness of generalization measurement through the train-test split and whether data leakage can be avoided. This is why the current research groups observations by the identity and recording data available in each dataset. Adding more components to the model alone cannot automatically address these issues.

A particular element may help in making the localization more effective even though it may not enhance the final task performance. While it keeps the posture data, it may discard the appearance data which is needed to match identities. Single-task models are therefore used as the benchmark before exploring how the tasks can be integrated into one framework.

1.2.3 Rationale for Jointly Studying the Three Tasks

The three tasks are indeed related, but each one is dependent on a particular type of visual information. BCS is dependent on the body morphology of the cow, behavior recognition is dependent on the activities of the cow, and Re-ID is dependent on the appearance of the cow. Due to their interrelatedness, integration is helpful, but due to their differences, the degree of sharing becomes a subject of study. Features associated with identity should continue to be useful regardless of changes in the activity of the cow, whereas behavior features should change when the activity changes.

Pretrained perception models make these comparisons possible without training every supporting model from the beginning. Promptable segmentation and transferable animal pose estimation can provide foreground and structural information [30], [36]. However, simply using these tools is not the contribution of the thesis. The contribution comes from testing how task-appropriate cattle-centered information affects the three tasks and how that information can be combined.

1.3 Problem Statement

This thesis investigates learning and sharing of visual representations for cattle BCS and behavior recognition and Re-ID without compromising information required by each task. Generic RGB features are a good starting point, but they can also confound useful animal information with background or environmental correlations. Another problem with heavy parameter sharing is that a shared representation must support tasks that may prefer different features.

For BCS, the model ought to acquire knowledge regarding body-condition data, rather than attributes that distinctly recognize the cow or its environment. The model requires spatial and temporal data without the necessity of associating a specific action with a particular place for behavior identification. In Re-ID, the model is expected to preserve an individual’s identification regardless of variations in camera perspective, backdrop, or temporal context. Consequently, satisfactory execution of a single task does not suffice to demonstrate that the same representation is applicable to all three jobs.

However, this research does not include combinations of related tasks such as disease type and disease severity. Through analysis of BCS, Behavior, and Re-ID simultaneously, it would be possible to know in what aspects similar features are needed and in what aspects it is required to process each task separately. The previous work on BCS based on anatomy, temporal Behavior recognition, metric-learning Re-ID, and selective multi-task learning gives different approaches for solving the problem. This paper combines previous work and investigates how cattle-centered visual information affects each task, how much information should be shared, and how the models perform under different recording conditions.

The primary research question is:

Are cattle-centric visual representations better than generic shared RGB representations in improving the robustness of body condition score, behavior recognition and cow re-identification and allowing a multi-task framework with less negative transfer?

To address this issue, we must analyze not only the task performance but also the visual information provided to the model, the sharing approach and the evaluation criteria.

1.4 Objectives

The primary objective is to develop and assess a multi-task deep learning framework for BCS, behavior recognition, and cow Re-ID using task-specific cattle-centric visual representations.

The first goal is to create evaluation protocols that preserve the relationships within each dataset. When reliable biological IDs are available, biological cow IDs are used. When they are not available, recording, passage, or session groups are used. This provides a safer basis for comparing models.

The secondary goal is to produce RGB single-task baseline models for the three tasks. These models provide a basis for assessment of the effect of changes in representation, and for the characterization of positive or negative transfer after integration.

The third objective of this research is investigation of cattle-centered visual information such as location of the cattle, foreground masks, anatomy/pose information, and viewpoint information. Temporal information will also be included for Behavior Recognition. The goal is to find out the combination suitable for each task while keeping the system feasible for training and evaluation.

Objective four will allow us to test heavy feature sharing vs a task-specific architecture based on the segregation of specific features or processing streams to perform certain tasks. This will allow us to ascertain whether task specialization allows for maintaining the sharing of information without hampering the efficiency of the tasks

performed. Objective five will allow us to test the robustness of the system under various image acquisition conditions as well as with similar external datasets.

The above objectives give rise to the following three research questions. Firstly, what is the impact of cattle-centric representation on the BCS, Behavior Recognition and Re-ID compared to the single task RGB model? Secondly, how can we incorporate the temporal and task-specific visual information into our framework?

1. How do cattle-centered representations affect BCS, behavior recognition, and Re-ID relative to their RGB single-task references?
2. How can temporal information and task-specific visual cues be incorporated while preserving the different requirements of condition assessment, activity recognition, and identity matching?
3. How does task-conditioned sharing compare with hard sharing in task performance, negative transfer, and robustness under domain change?

1.5 Methodology in Brief

In the present research, the efficiency of task-conditioned sharing is examined in comparison to hard sharing regarding task outcomes, negative transfer, and robustness in case of changes in domain conditions. The experimental comparison design is used. First of all, the data pre-processing stage identifies the labels and the groups which will be tested. ScienceDB will serve as the primary source of the BCS dataset; CVB and Kaggle Beef datasets will be used for the Behavior dataset; and SideViewCows2026 for the Re-ID dataset. Various datasets were selected as each task is based on different observations and labels.

Single-task RGB models are used as reference models. Their cattle-centered versions use localization and binary-mask guidance, while the behavior model also uses temporal aggregation. Pose, anatomy, and viewpoint cues are studied separately instead of assuming that they will transfer well across datasets. Upstream perception is checked before its output is used in a downstream model.

During multi-task experiments, various methods of integration will be compared to the single-task cattle-centric models on the same held-out test sets. The monolithic model with hard sharing will be tested in order to estimate the performance of full parameter sharing between the tasks. A modular task-private model with separate residual bottleneck adapters will also be tested.

PCGrad is utilized in another multi-task model to test the influence of conflict resolution in the gradients. The architecture as well as the total number of parameters remain the same as the hard-shared one. Negative transfer is assessed separately for each of the tasks through using the held-out dataset. As for BCS, prediction errors are taken into account. Class-sensitive results are evaluated in case of Behavior, whereas open set retrieval is used for Re-ID. Some other compatible datasets could be tested in the future for cross-environment evaluation, however, they remain outside the main benchmark evaluation.

In the chapter 4, it discuss about model settings, training methods, experimental controls, as well as data divisions. And in the chapter 5, it discuss about the result as well as the interpretation of the result. These experiments show what information each task needs and what happens when the tasks learn together.

1.6 Scopes and Challenges

The framework covers three visual tasks, BCS, behavior recognition and individual cow identification (Re-ID). The thesis does not cover lameness detection. RGB imagery is the common visual input. Depth and other external data are only used when their role and label compatibility are clear. The presented framework is a research contribution to the monitoring of cattle and not a replacement of the veterinary assessment.

The tasks and datasets give different type of supervision. BCS labels vary in coverage and scoring. The sort of behavior depends on the position and the activity. certain categories are present in certain data only. Re-identification requires a well defined division between training and assessment datasets, and robust bovine identity data. ScienceDB does not do biological cow-disjoint assessment, but rather burst-group-disjoint and sequence-safe evaluation. These differences restrict the assertions that may be made from each experiment.

Moreover, there are some image conditions like viewpoint, occlusion, scale, and lighting that can affect the amount of information in the picture. Foreground processing could eliminate the unnecessary background; however, sometimes valuable information could be excluded from the scene as well. Pose estimation provides structural information about the body, though it may lack some of the necessary anatomical information for BCS. That is why the effectiveness of each of the components is determined experimentally rather than being presumed beforehand.

In addition to this, the structure of the research is dependent on the computational capabilities. In this case, the use of the pre-trained models and reusability of the pre-processing methods and limitation of the comparison scope are applied not to create an excessively big model. The descriptions of the previous work and the research gap are presented in Chapter 2. The implications, requirements and limitations are discussed in Chapter 3. The description of the methodology and experimental analysis are provided in Chapter 4 and Chapter 5 correspondingly. The conclusion is made in Chapter 6.

The completed research can be associated with the leakage-aware evaluation, single-task comparisons with the cattle-centered representations and the multi-task experiment with the control group. The single task experiment revealed that the effect of cattle-centered representations is different for various tasks. Hard-sharing led to a bad performance in multi-task experiments; however, other architecture and optimization techniques have helped to reduce some of these problems. The acquired results support the idea of a unified model, which will incorporate some information sharing about cattle, but taking into consideration different representation and optimization of tasks.

Chapter 2

Literature Review

2.1 Preliminaries

In the case of a unified cattle monitoring system, it is necessary to utilize various types of visual cues jointly. In particular, body condition relies mostly on body structure, whereas behavior relies on body posture and motion. On the contrary, identity is defined by a set of visual features, which allow distinguishing between individuals. That is why the thesis merges concepts from livestock monitoring, visual representation learning and multi-task learning.

This chapter will provide an overview of the existing studies of different types of visual cues and evaluation strategies for them. First of all, the basic terminology regarding the three tasks will be discussed. Then, cattle specific models, foreground processing, anatomy and pose, temporal information and task specific feature sharing will be analyzed. Finally, these concepts will be used to describe the research gap addressed in this thesis.

2.1.1 Visual Representations and Transfer Learning

The visual representation is the set of features that the model uses to represent an image or a video frame sequence. Previously developed recognition algorithms relied on manually selected features. Neural networks trained on a dataset learn relevant features without any design effort on the part of engineers. In the case of convolutional neural networks, local primitive features are composed to produce more sophisticated ones gradually. The visual representations fed to the predictor define the type of information the model uses to make predictions.

Reusable image feature extractors have enabled this step in specialized domains. ResNet [4] uses residual links to facilitate the training of deep networks. EfficientNet [18] optimizes the scaling of network depth, width, and input size together. Even though they solve different problems in their designs, both architectures may be useful starting points for transfer learning. In this particular instance, instead of using the machine learning algorithms to learn all the visual features of the cattle based on the limited dataset available, the pretrained machine could be adapted for this particular use.

Transfer learning and cattle-centered representation learning are not the same in terms of functionality. Transfer learning focuses on the source of initial weights of the model. Cattle-centered learning gives us a clue as to what kind of input data was

fed to the model. The same pre-trained encoder may be used with different types of input data such as full image, crop of the cow, masked cow, and anatomically detailed images. That is why the encoder will stay the same regardless of the changes in the input data. This difference is important when comparing why one representation performs better than another.

2.1.2 Prediction Tasks and Their Visual Requirements

There are different approaches for predicting BCS, which are classification, ordinal prediction, and regression. In classification, all the scores can be treated as individual classes. In ordinal prediction, order is significant since the prediction of a neighboring score is a less significant error compared to the one which is distant. However, regression just tries to predict the numeric output. CORAL is one example of an algorithm designed for the ordered labels [19]. This algorithm may teach the classifier about the order of BCS, but it will not choose the visual features of the cow.

In the case of behavior recognition, a class is actually an activity that is not measured by numbers or scores. One frame can be able to inform us about the posture of a cow, whereas multiple frames can show how the cow moves over time. It is important since there are activities that might be quite similar in one frame only, for example, standing and walking. which can look almost the same in a single frame. So, the model should use a suitable amount of time for the detection task.

Identification and Re-ID are two distinct tasks. Closed set classifier identifies the cows seen in the training dataset. On the other hand, the query image is compared with the gallery images based on the embedding learned in Re-ID. It becomes possible for the system to be tested on cows unseen during the training phase. The good Re-ID embedding will classify images of the same cow into one cluster and images of the other cows into different clusters. The significance of this technique in cattle research lies in the changing herd of cattle [24].

2.1.3 Multi-Task Learning and Negative Transfer

Multi-task learning entails the training of different tasks in the same model where particular parts of the model are common and the training objective is minimizing the total loss of all the tasks,

$$\mathcal{L}_{\text{MTL}} = \sum_{t=1}^T \lambda_t \mathcal{L}_t, \quad (2.1)$$

where in this equation, $\mathcal{L}_t$ is the loss of task t , and λ_t determines the weight of the loss of task t . This equation calculates the total loss; however, but there is no information about what amount of the network should be shared. A neural network model can share all layers, a few layers, or exchange information between tasks.

In the survey paper by Crawshaw [20], MTL is considered in the context of information sharing, task-specific components, and optimization. MTL could be advantageous if one task is providing some useful information for another task. However, joint training could be harmful to a specific task; that is, the effect is referred to as negative transfer. It might occur due to the inappropriateness of information shared between tasks, conflicting task gradients, and unbalanced training losses. It

is critical to distinguish information sharing and training for the purpose of the thesis.

2.2 Review of Existing Research

2.2.1 From Automated Observations to Integrated Livestock Monitoring

The development of the livestock monitoring technology consists of moving from individual measurements towards integrated systems of sensing, prediction, and decision-making. One can see this in the reviews that were written about computer vision as well as in the precision livestock literature in general. Within the scope of this particular question, the work of Antognoli et al. [38] has to do with computer vision in dairy farming, whereas the research by Zin and Tin [51] refers to computer vision in livestock in general. It shows that image recognition is not the only part of livestock monitoring.

First, the image should give a proper look at the animals themselves. Palma et al. [42] revealed that predictive analysis is applied more frequently than prescriptive decision support in the dairy literature they examined. Similarly, Sani et al. [48] pointed out that it is challenging to employ precision livestock technologies on-farm. This means that mere data collection or enhancing the performance of one classifier is not sufficient. The results need to be tracked, as well as associated with the right animal and farm condition.

Sensor-based and vision-based systems yield different types of information. As Lee et al. [47] indicated, biosensors enable continuous information acquisition about individual animals, whereas cameras make it possible to capture the appearance and behavior of the animals directly. The importance of vision for this thesis comes from the fact that information about the body shape, posture, movements, and coat patterns is available through images and videos. All the three tasks utilize the same type of information source, namely visual information, though they require different information out of it.

There is no need to have exactly the same components for all the tasks in the unified cattle monitoring system. Some components, like cameras, preprocessing, and shared features can be the same, but others will be different for each particular task. Thus, apart from integration of different tasks' predictions, the process includes identification of those features that should be used for all tasks and those that should stay specific.

2.2.2 Body Condition Scoring: Anatomy, Image Geometry, and Ordered Prediction

Automation of BCS is highly dependent on the cow's anatomy. Edmonson et al. [1] introduced the scoring index, whereas Ferguson et al. [2] explored the connection between the animal's anatomy and certain BCS values. Regarding the computer vision aspect, body mass of the cow alone is not enough as an indicator. Indeed, there should be considered such parameters as tissue covering, body shape, and crucial body parts. Hence, camera angles and the body parts themselves become very important.

Another approach uses depth images for better representation of the body shape of a cow. In particular, Rodriguez Alvarez et al. [13] employ convolutional neural networks along with depth images and validate the prediction of BCS by means of various error thresholds. This approach relies on the combination of body geometry and features rather than the handcrafted features alone. The paper also proves that exact accuracy and tolerance-based accuracy have to be evaluated separately because of the fact that BCS scores are ordinal.

Another methodology makes use of ordinary RGB images to detect visual cues associated with body condition. In the case of Liu et al. [41], most of the feature extraction takes place in the tail area and is done using the EfficientNet-B0 architecture with channel and spatial attention. Their work involves the selection of a specific body part and an efficient network architecture, while distillation assists in making the model size smaller. The fact that BCS prediction can be tailored through a combination of the specific body part and architecture has been illustrated.

Camera view influences the information that can be obtained by the model as well. Yao et al. [49] select side-view pictures of the cattle using a ConvNeXt-based regression model and cow-level cross-validation. One method is focused on the tail of the animal; therefore, it observes just a small portion of the body, whereas another method selects side-view pictures and considers a bigger part of the cow’s body. Both methods are different not only in their models but also in the parts of the cow they observe.

The research done on cattle measurement also reveals that there is a need for distinction between image size and true physical measurements. The authors Guzhva et al. [46] came up with a system called PickAMoo, which uses smartphone images, LiDAR scaling, and Mask R-CNN segmentation in order to estimate weight. Even though the purpose of estimating the weight is different from estimating BCS, this method is relevant since the estimation is done when the animal has been separated from the image, and pixels are related to physical measurements.

The common factor among all these papers can be seen in three aspects that make good components for BCS prediction. First is the use of images that have proper features of the body and its orientation. Second is the use of an appropriate technique to extract such information, for example, cropping, body contour, depth map, or attention. Third is selecting an appropriate prediction technique that is suited to ordered BCS. CORAL can handle ordering of output classes [19], but it will not perform well if the input class does not have appropriate condition features. Thus, body-shape information will be considered as part of the representation, and ordinal prediction as the model choice.

2.2.3 Behavior Recognition: Posture, Motion, and Environmental Context

Behavior classes can differ in both appearance and duration. Lying may often be recognized from body posture, while walking is better described by movement. Feeding and drinking also involve interaction with a resource such as a trough. Because of this, behavior research uses different amounts of spatial and temporal information, and a single-frame classifier cannot describe every type of activity equally well.

Image-based behavior recognition shows that posture can already provide useful information. Asim et al. [44] use overhead-camera images and object-detection models

to recognize activities including standing, lying, grazing, and estrus-related behavior. Their method is vision-based rather than wearable-sensor based. It shows one way to connect animal localization and activity labels in a frame. A sequence model can then add information about how those visible postures change over time.

Video datasets extend the task beyond one frame. CBVD-5, introduced by Li et al. [33], contains standing, lying, foraging, rumination, and drinking. CVB, introduced by Zia et al. [32], provides densely sampled behavior clips from an open-pasture setting and includes a SlowFast benchmark. The datasets have different environments and behavior labels, but both show that behavior recognition depends on the animal and on how the activity is defined and observed over time.

MmCows [35] is the fusion of synchronized sensor data, information about the cow and video data. The annotations of MmCows include identity, location and behavior of the cows observed from different viewpoints. One can see whether behavioral predictions stay consistent across viewpoints and individual cows. Although a video dataset with tracking IDs is still useful in terms of motion study, it cannot provide the same level of biological identity information.

These demonstrate two different aspects of behavioral data: temporal information and identity consistency. Video in high density helps with motion analysis in short term. Identity of the cows helps with the test being done on different cows and across time. The one does not substitute for the other. This means that a study that requires both needs to factor in both and not just choose the one with many images.

Contextual background information can also be useful and detrimental. While a dip or feeding barrier could aid in learning about feeding/drinking, the problem is that it might learn that one task tends to happen in one camera quadrant. While the removal of the entire background will make the shortcut less obvious, it will also take out some interaction information. It is not about whether the context can be seen as positive or negative. What matters is the amount of context that is helpful to comprehend the activity and its ability to remain relevant despite changes in the environment.

This is linked to the cow-focused thesis in which the cow is still the key element of the visual reference and where posture, movement and context relationship can all be used if it is necessary. It also allows for the use of a class-based assessment – a model can perform well on common rest behaviours and poorly on rare or more dynamic ones. This can be obscured by the overall accuracy if class frequency is correlated with source.

2.2.4 Individual Identification and Re-ID: From Coat Patterns to Cross-Setting Recognition

The visual identification of cattle requires certain characteristics that differ from one animal to another. Andrew et al. [6] use the technique of local coat pattern matching through RGB-D images. This technique provides us with an example of how the useful characteristic of identity is made explicit: local patterns are able to identify Holstein-Friesian cattle. This technique also indicates an important drawback that remains relevant for today’s models. The identity-related feature must be visible in both images that are being compared.

Deep metric learning changes how these comparisons are learned. Andrew et al. [24]

use an embedding-based method with a combined classification and reciprocal-triplet objective. Instead of only using a classifier for known cows, the embedding supports similarity-based matching. This allows the feature space to be tested on cows that were not used to train the encoder. The goal therefore changes from memorizing fixed identity labels to learning features that can separate animals more generally. Face-based identification uses a different body region. Weng et al. [28] use a two-branch convolutional network to combine cattle-face information. Compared with coat-based methods, this focuses on facial differences between cows. Which approach is more useful depends on the camera view. A clear face may be hard to obtain from an overhead or rear view, while a useful coat pattern may be hidden by occlusion or missing from a tight face crop. Camera position and visible body coverage are therefore important parts of Re-ID.

Another line of work reduces the amount of manual identity labeling needed in video. Gao et al. [25] use the Cows2021 dataset to study self-supervision from tracking-based observations and normalized cow views. Yu et al. [43] extend this idea to several cameras on a working farm with MultiCamCows2024. These methods use the continuity of video to connect images that are likely to show the same cow. A short tracklet can therefore help the model learn appearance consistency without requiring a manually assigned cow ID for every frame.

Nevertheless, tracking and Re-ID are still different tasks. While tracking maintains association of an animal when it stays continuously visible, re-identification must restore this association when there is a gap in time or change in viewing conditions. Information about tracking may be used for training of the Re-ID model, but an id of the temporary tracking does not correspond to biological ID of the cow. Multi-camera and cross-setting assessment would be helpful since it examines bigger shifts than a continuous video sequence.

The other problem is identification over a prolonged period of time. The BECA database [50] proposed by Zhang et al. contains not only a big cattle population but also a longitudinal setting. This setting differs from the short-term identification of dairy cows which can possess unique and distinctive patterns. The appearance of the animal can change due to the size, position, and body parts depicted in the image. The bigger the cattle population, the more similar the animals can become to each other. Therefore, a Re-ID system should not only be assessed in terms of the identification of trained cattle population but also in terms of others.

Similarly, foreground processing and anatomy processing have to take into account the need to preserve the data needed for identification purposes. While deleting the background may force the model to concentrate on the cow, deleting too much of it by reducing it to just a silhouette or skeleton may lead to loss of critical coat features. Retaining the full environment will cause the opposite effect by teaching the model the usual position of the cow. Consequently, Re-ID will require a selective representation technique that minimizes environmental shortcuts but maintains cow-specific characteristics.

2.2.5 Localization and Segmentation as Representation Design

The localization determines the cow on which the model needs to be applied. This becomes very crucial when there are multiple cows in the same image since the

predicted class or behavior belongs to that particular cow only. The bounding box provides an effective ROI, but also maintains some context within the image. Image cropping can normalize the scales and enable the encoder to focus, but alters the body-background ratio.

There are several modern object detectors that can help to locate this region. RT-DETR developed by Zhao et al. [37] uses an end-to-end transformer detector optimized for speed-accuracy trade-off. Instance segmentation approaches are richer than bounding box. Mask R-CNN extends the detection network with the mask branch and thus makes it predict not only the position but also object specific mask [7]. To be more precise, the bounding box says where the cow is, while the mask defines its pixels.

Promptable segmentation decouples the process of selecting the target from creating its mask. Promptable segmentation model, Segment Anything by Kirillov et al. [30], is capable of generating masks for prompts such as points and boxes. Promptable segmentation in SAM 2, being memory-based, is applicable to both images and video [34]. This implies that in a cattle pipeline, for instance, we can first detect the cow, and then prompt the segmentation model to generate the mask, without the need to train a new cattle-specific segmentation model from scratch. The final mask will depend upon how the subject was chosen and prompted.

Supervision becomes more straightforward in cattle-specific datasets. CattleEye-View [31] is a dataset with annotations from a top-view for detection, counting, tracking, pose estimation, and instance segmentation tasks. As multiple annotation types describe the same scenes, the data set is relevant to cattle perception. Nevertheless, these tasks of cattle perception are different from predicting BCS, Behavior, and Re-ID. They are mentioned in this thesis to characterize the cow, yet, they do not provide proof of the representation being suitable for all other tasks.

Recent cattle recognition works relate foreground detection and cattle identity learning. CATR-Net [40] uses segmentation and cattle face recognition techniques in combination with edge-based segmentation and attention-based recognition. It associates the process of separating the cattle from the background with identity-related characteristics. In addition to previous approaches based on coat appearance and face recognition, this illustrates the relevance of the cattle-centered recognition as an active research area. The question remains open whether foreground-oriented processing is beneficial for tasks depending more on body shape or movement as well.

Segmentation and feature attention are also different. CBAM changes the importance of channel and spatial features inside a network [14], while a foreground mask changes or weights the image content using an estimated object boundary. Attention can focus on useful regions while still keeping the full image. Masking can reduce background information before the main encoder. The two methods may work together, but they are not the same intervention.

Masking could also be applied with different degrees of strength. With hard foreground masking, all content outside of the estimated boundary is eliminated. The softer form of masking will lessen the effect of the background while retaining some context along with uncertain boundary information. It is a matter of selecting the appropriate approach for the particular task. Useful body contours are required for BCS, context is needed for behavior, and coat information for Re-ID. Therefore, the quality of a mask should be assessed together with the performance of the

downstream task, rather than only by how clean the mask appears.

2.2.6 Pose, Anatomy, and Viewpoint

Pose estimation represents an animal using the positions of selected body parts. DeepLabCut shows that user-defined landmarks can be learned with transfer learning from a relatively small amount of annotation [12]. This changes the input from dense appearance information to a structured description of body configuration. Such information can support movement analysis and can also make some parts of the model input easier to understand.

This is precisely what SuperAnimal [36] does by fusing various animal poses datasets to train models that will be applicable to different animals. SuperAnimal’s training strategy is flexible enough to incorporate differences in landmark definition, and it can either be directly applied or further modified. This is beneficial for cattle studies as it makes it possible not to start building a new pose dataset from scratch. Another study, which gives a more specific example is CattleEyeView [31], where a top-down approach was chosen specifically for cattle.

Overall, these studies provide an example of how structural information can be obtained in two ways – through transfer learning using different animal datasets and through direct training based on a cattle-specific dataset. Also, the definition of landmarks plays an important role. The skeletal information can be gathered from the head, the body, and legs, but BCS requires more. Thus, the BCS features that were investigated by Ferguson et al. [2] include much more than just positions of joints. Pose and body configuration are related, but these are different representations of the data. The BCS may require both the structural coordinates and visual features while Behavior will probably rely on the changes in the coordinates in time. Viewpoint impacts the visibility of body parts and appearance of the cow in the image. The rear, side, and top views may represent different body configurations despite the fact that the condition of the cow remains the same. This point is important for the task of Re-ID as the same cow must be identified regardless of view. This point is also critical for BCS as some informative body parts might be invisible from some viewpoints. Viewpoint corresponds to the direction of the cow in relation to the camera. It is not the same as camera identity.

The experiment conducted by Grolleau et al. [45] used synthetic cows created using fixed camera views. The advantage of such images is that, knowing the angles of view, one can analyze the effect of geometric variations independent of the changes which happen in reality. Moreover, Grolleau et al. [45] tried to see if the features extracted by learning from synthetic examples can be transferred to real cattle.

Synthetic images are good for viewpoint annotation, while real images test how well the features extracted are robust against changes in the appearance and environment. The main idea behind this approach is that the model should be invariant to some kinds of changes and sensitive to others, depending on the task. In Re-ID, it is required that the representation be invariant to any viewpoint variations of the same cow. On the other hand, in Behavior recognition, the variation in posture is an important element and should affect the model. In BCS, variation in scale does not preclude the usage of informative body shape cues. This is the reason why the thesis uses task-conditioned structured information instead of using the same invariant representation in all tasks.

Table 2.1: Visual information and task requirements synthesized from the reviewed literature. The entries describe design considerations, not measured outcomes of the proposed system.

Visual cue	BCS	Behavior	Re-ID
Foreground appearance	Body contours and local condition cues	Posture and interaction details	Coat markings and individual morphology
Pose and anatomy	Region alignment; joint locations alone may omit condition cues	Body configuration and its changes	Possible alignment aid; limited identity texture
Viewpoint	Visibility of condition-related regions	Interpretation of projected posture	Correspondence across views
Temporal context	Repeated observations of condition	Motion, duration, and transitions	Appearance continuity across observations
Scene context	Often secondary to body evidence	Potentially useful near feed or water resources	Possible background-dependent matching

2.2.7 Temporal Representations for Activity Recognition

The modeling of the temporal structure aids in detecting changes in time. Different types of temporal structure also depend on the type of representation used. RGB sequences provide information on the appearance and posture of the cow whereas keypoint sequences provide information on the body structure. Video models can detect information on space and time simultaneously. However, the two models differ in terms of retaining information as well as computing and labeling processes. SlowFast network explicitly separates the aspects of appearance and motion using two streams with different frame rates [16], where the first stream concentrates on spatial information and the second captures faster changes. SlowFast network is also used as the benchmarking technique in the CVB application [32], making connection between general video recognition technique and behavior of cattle. It provides a stronger motivation for temporal modeling approach other than simple assumption of more being better.

A temporal convolution network may also be used. In comparing convolutional and recurrent sequence models, Bai et al. [8] found that temporal convolutions have shown promise in several sequence tasks. These models leverage temporal properties without needing to include a recurrent hidden state at each step. In relation to cattle behavior, this provides justification for using a light temporal network in place of image features when many visualizations must be compared.

Another way to encode motion is to consider body landmark relations over time. ST-GCN by Yan et al. [15] is a model that uses graphs to represent the skeleton of a moving human being with connections between nodes in space and in time. This approach is applicable to the problem of interest due to modeling relations between body parts rather than modeling coordinates separately. Still, its use in the case of cattle depends on having correct landmarks definition and accurate estimation

of poses. Therefore, the pose sequence should be seen as a different representation category and not a substitution for RGB video.

The approaches prove that different temporal inputs address different problems. Taking the average of several frame features provides the model with more observable appearance, while a temporal model can also make use of the sequence of those frames. Pose features concentrate on body pose, while dense video contains more motion and contextual information. This choice is based on the type of action and sampling period. Hence, this thesis makes use of temporal modeling as a representation problem rather than scaling up the model.

2.2.8 Shortcut Learning, Domain Shift, and Evaluation

Good performance in a certain environment does not necessarily mean good performance in another environment. Geirhos et al. demonstrated how deep learning representations can base themselves on simple associations which may be vulnerable in new environments. The matter becomes even more critical in case the object and the background of interest recur often. For instance, images of cattle can always include the same pen, illumination, and perspective, which may make it easy for the algorithm to take advantage of some scene characteristics.

It is possible to test the dependence of the algorithm on the background directly. Xiao et al. [27] vary the information about the foreground and background in order to find out their role in image identification. The results of their experiment show that the evaluation should not be restricted to using the same dataset only.

In case of cattle-oriented learning, the major point will be whether the algorithm continues paying attention to the cow under varying parts of the scene. Foreground processing and changes in camera, setting, or dataset can also be used to test this robustness. According to Beery et al. [9], there is an enormous discrepancy in terms of model performance between familiar camera-trap sites and unfamiliar camera-trap sites. Camera traps for wild animals are distinct from livestock cameras; however, they both may have the same fixed background repeated multiple times. The WILDS benchmark [26] also considers this problem using the benchmarks which feature natural distribution shift in terms of wildlife and other cases.

Thus, models need to be tested under various conditions, rather than considering all test images as challenging. The domain shift is not confined only to background change. The domain shift can include such factors as different cows, new breeds, new body appearances, new camera heights, as well as different label definitions. Re-ID studies [43], [50] analyze some of these variations via multi-camera and long term settings. It is also important to take into account the image condition as well as whether the labels have the same meaning across different datasets.

It is also critical to split the evaluation data appropriately. As Kapoor and Narayanan [29] demonstrated, data leakage can lead to an erroneous estimation of performance for machine learning approaches. For video datasets, it is possible that two images might still refer to the same continuous sequence of events. Consequently, random splitting of images would mean that very similar scenarios would be included both in training and test sets. The correct grouping of the dataset would be a more effective approach in such cases. The choice of the correct grouping unit will depend on the type of evaluation. For unseen-cow evaluation, the correct cow ID is needed, and grouping by recording avoids another kind of overlapping.

In this sense, the design of the evaluation procedure is part of the representation problem. The mask will seem relevant in one split since the model is dealing with an easy identity and localization task. The use of multiple datasets will bring variation visually while one dataset is still linked to one class. Evaluation and training should be aligned with what is claimed in this case. This brings more validity to testing whether the use of cattle information helps with transfer rather than improving performance within one known data set.

2.2.9 Multi-Task Learning: Selective Sharing and Negative Transfer

The use of multitask learning (MTL) allows the combining of the three models into one system. The most basic design uses a single encoder and different heads for each task. This helps in saving computational resources and encourages the use of common features. However, the use of MTL requires more than the fact that there are three inputs containing cattle. Crawshaw [20] notes that architecture, task relationships, and optimization all have roles in MTL performance. Thus, the sharing of the resources should have something more compelling than just the fact that the three inputs have cattle.

Some of the methods used in MTL attempt to ensure the balance of task updates. In Kendall et al. [11], loss weighting is done using task uncertainty. The other technique is called GradNorm, which uses gradients for modifying the task weight depending on learning pace [10]. These techniques seek to prevent dominance of a particular task during joint learning due to either loss magnitude or pace. They control the strength of each task’s update to the network. However, they do not select features to be shared visually.

PCGrad solves a completely different aspect of the problem through the modification of gradients during conflicting task updates [23]. Thus, gradient interaction becomes a component of the optimization process. Modification of the magnitude of an update and modification of a conflicting direction are two separate things, and therefore, PCGrad cannot be equated to loss weighting. Both these techniques remain within a selected architecture, thus allowing researchers to explore optimization as one of the causes of negative transfer.

Another approach alters feature sharing between tasks. Cross-Stitch Networks [5] learn how to share activations between task-specific networks. The Multi-Task Attention Network presented by Liu et al. [17] employs task-specific attention for selecting the relevant data from common features. It implies that all the tasks do not necessarily have to follow the same feature pathway. Such an approach can prove to be helpful if the tasks are somehow related but at the same time, require the extraction of different information, like cattle body structure, behavior, and identity.

Task clustering can be evaluated instead of being predetermined beforehand. Standley et al. [22] considered the question of the most suitable task clustering if the number of resources is limited. The research conducted by the authors showed that the relation between the tasks cannot be defined just based on the names or descriptions of the tasks. Two tasks may share early visual features but require different processing later, and one task grouping may help one task while making another worse. Good models for each single task need to be achieved before testing how

Table 2.2: Complementary approaches to managing multi-task interaction. The methods address different aspects of joint learning and therefore support different interpretations of an improvement.

Approach	What changes	Question addressed
Uncertainty weighting [11]	Relative contributions of task losses	How should losses with different uncertainty be balanced?
GradNorm [10]	Adaptive task weights using gradients	How should task learning rates be balanced?
PCGrad [23]	Interaction of conflicting updates	Can harmful gradient interference be reduced?
Cross-stitch sharing [5]	Exchange between task-specific features	Which feature combinations should be shared?
MTAN [17]	Task-dependent selection of common features	Which shared information is useful to each output?

good a joint model is.

Previous research works on animal vision have demonstrated both benefits and drawbacks of multitask learning. The SAAM-VetNet [39] is designed for disease detection and classification of its severity via attention-based multitask learning. Also, CATR-Net [40] joins cattle face segmentation and recognition. It is shown that multitask learning and perception-based recognition could be helpful in the case of animal vision. Nevertheless, the relationship between tasks in the previous studies differs significantly from the relationship in the present thesis. Segmentation and face recognition are oriented towards one visible object, whereas the disease type and its severity rely on similar local symptoms. Meanwhile, Body Condition Score, Behavior, and Re-ID require more diverse types of information.

Re-ID relies on the features which stay constant across the images of the same cow. Behavior recognition requires a response when the same cow changes its activities. Body Condition Score needs information about body shape without replacing condition information with cow-specific appearance. The common encoder would be possible for all three tasks in case it maintains the distinction between them. That is why the thesis contrasts hard sharing with task-conditioned or partially independent encoding.

The application of different datasets to the tasks poses another challenge. Joint training involves the perception of differences in perspective, composition, and style of annotation alongside the difference in tasks' goals. Consequently, some distinctions between tasks may be related to the used datasets. The fair comparison would involve maintenance of each task's evaluation protocol and analysis of results of every task separately. It would help to reveal whether the selective sharing really improves the model or only increases its size.

2.3 Summary of Key Findings

Therefore, the research problem is not limited by merging of three cattle classifiers. BCS, Behavior, and Re-ID need different types of visual information. BCS based

on anatomy, Behavior models based on time, and Re-ID based on metric learning have different solutions for cattle monitoring problems. Each of them emphasizes different features of the cows.

2.3.1 A Common Animal, but Different Information Requirements

Thus, there should be a unified approach that considers the differences of all of them. BCS is based on the degree of body covering, body shape, and BCS score. Behavior Recognition depends on the movement, duration, and interactions with the environment. Re-ID depends on the consistent appearance of the cow in the various observation cases. This may be complementary in some cases, however, it may also lead to conflict situations. For example, the absence of individual textures can be helpful in avoiding identity-based BCS shortcuts, but it is required in case of Re-ID task. The presence of scene information may help in detecting feeding behavior, but it can create a shortcut for Re-ID.

Different approaches exist previously for different cattle monitoring tasks. Research gap #1 is that it remains unclear how the above-mentioned options will impact BCS, Behavior, and Re-ID, as well as what will be the performance of cattle-centric representations against RGB representations under the consideration of all three tasks in one unified framework.

2.3.2 Representation Quality Must Be Connected to Robust Evaluation

Perception-based models can provide masks, landmarks, and viewpoint data; however, the generation of the output is not enough in terms of ensuring that the resulting representation contains all the necessary data for the considered tasks depending on changing conditions. In the case of BCS, these conditions involve cattle’s body anatomy and conditions of capturing an image. In the case of Behavior, they involve cattle’s posture and movements as well as changes in the scenery. And in the case of Re-ID, they involve different perspectives of one and the same cow in different locations or at different times.

This creates research gap #2, where changes in representation need to be tested under evaluation conditions that match the expected type of change. Foreground-oriented models need to be assessed using cross-setting evaluations and class-wise results and not just an overall figure. Temporal models require to be evaluated using cross-temporal sampling of observations. Thus, this thesis investigates the relationship between visual data, task demands, and generalization settings.

2.3.3 Selective Sharing as the Basis of the Unified Framework

Multi-task learning studies consider two problems, that is the problem of balancing updates from different tasks and sharing information between different tasks. Early experiments in this thesis find that this distinction is significant in cattle monitoring applications. Rather than assuming that high feature sharing is desirable, first, we investigate the needs of each task individually using single-task models.

Then different multi-task learning schemes are considered and compared. The study seeks to make use of the similarity in the cattle but pays varying attention to body shape, behavior, and individuality. The main comparison will be between the generic RGB baselines and hard-sharing approaches. The study will focus more on task performance and negative transfer. Moreover, which features can be shared, which cannot be shared unless task-specific processing is done, and how they influence the integration will be investigated.

The literature review above will serve as a foundation for the subsequent chapters. Task constraints are provided in Chapter 3. Data protocols, representation, and model comparison will be discussed in Chapter 4. Chapter 5 will address the research questions using collected data.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

This system is intended to deliver three predictions. First of all, it should predict the condition of the animal. Second, it should determine the type of activity that the cow is engaged in. Third, it should provide identity representation for Re-ID. This solution is intended to be used as a prototype. Therefore it should be able to generate the predictions and ensure their correct evaluation this system relies on labeled data, appropriate splitting of the data set handling failures, correct metrics for evaluation and reproducibility of the environment for BCS. The prediction should belong to the BCS interval for behavior. The prediction should belong to one of the five classes for Re-ID. The identities of the training cows gallery and query images should be defined.

Table 3.1: Research requirements and the evidence needed for acceptance.

ID	Requirement	Evidence required
R1	BCS output with explicit units	Physical-score mapping, output support and saved metrics.
R2	Target-cow Behavior output	Taxonomy, crop/cache integrity and class-wise results.
R3	Re-ID under a named protocol	Identity partition, query/gallery definition and retrieval artifact.
R4	Evaluated integrated model	Matched single-task / E1 / E3 / E4 comparisons and documented sharing / optimization boundaries.
R5	Provenance-aware evaluation	Grouped splits, versioned checks and limitations.
R6	Reproducible execution	Executed settings, seed, revision, hashes and checkpoint.
R7	Transparent failures	Missing-input/perception counts; no silent substitutions.
R8	Feasible implementation	Measured resources and maintainable dependencies.

R4 relates to building a multi-task model from the combination of all three tasks. The requirement is supposed to provide a clear explanation of what components will be shared by the model and what will be task-specific. The main non-functional requirements are reproducibility, resistance to data leakage, maintainability, proper failure reporting, and execution feasibility. All dataset manifests, scripts, metrics and configurations are under version control, but raw datasets and large checkpoints are not. Therefore, reproducing an experiment presupposes having both the logged configuration and the right dataset.

There are some additional requirements related to scientific validity. For example, test data cannot be used for choosing checkpoints. In case some samples are dropped due to perception failures, there should be either matched samples for comparison or a clear coverage indication. In case some task labels are missing, they are supposed to be dealt with using task-specific batches or any other approach that does not rely on the missing labels. The missing labels should not be considered negative ones. Each evaluation needs to align with the nature of data splitting that was done. Examples are bars protection, source videos, recording sessions, or individual cow identifiers. There exists a standard visual monitoring system that could eat in linking information related to body condition and behave with the individual cows. It will eat in early detection of anomalies and hence avoid the TDS task of analyzing many videos. However, this research leaked farmer, interviews and field implementation. It did not also quantify any of the possible benefits, such as lab, savings, productivity, improvements, efficacy, and animal welfare. Those are future benefits, and not what was measured in the study. This computational cost of prediction is only one aspect of the total computational cost. Others are image pre-processing model training, data storage and data transfer

3.2 Societal Impact

The model should assist human decision making, not replace it. Poor identification matching can lead to observations being assigned to the wrong cow, and an inaccurate BCS or behavior forecast can cause to a delay in attention. A realistic system has to save the original image, the time, the confidence or proof of retrieval and a way for a person to judge the output. There still has to be good professional judgment in veterinary and welfare decisions.

Conditions at the farm will be an important aspect when using the system. Aspects like camera placement, accommodation, breeds, lighting, internet, and staff working procedure may vary from farm to farm. Due to this, the current findings cannot confirm the system’s success in all areas including Bangladesh. Tests must be done by people in relative fields before claiming that the system works in real world.

Agricultural images may depict people, arrangements, or personal information. There is no guarantee in the license to utilize a database in research that the images may be distributed or used commercially. Privacy, ownership, and licensing regulations must be considered for each database and each potential field use.

3.3 Environmental Impact

. A large model detector/segmentation model can impact the performance of the smaller model used later on in the pipeline. Hence, such an impact needs to be properly accounted for.

In order to avoid duplicating efforts, the pipeline was inspected prior to running the experiment. In addition, reuse of the previously validated output was performed in order to avoid any unnecessary redundancy. However, it needs to be noted that no measurements of the energy or carbon footprint were taken in this study. Thus, no actual savings in carbon emissions can be claimed by this study.

Effective tracking of the cattle can make farms more efficient in the future. Nevertheless, no feed savings, reduced methane or illness rates were measured in our experiment. Such results will depend on farmers' application of the predictions and will require another lengthy experiment.

3.4 Ethical Issues

Another important aspect in regard to ethical issues is distinguishing facts from hypotheses. The reason why ScienceDB is not cow-disjoint is that there is no cow ID that is scientifically proven in biology. In addition, the main Behavior analysis cannot be referred to as unseen cow analysis. Ground-truth masks are needed for Oracle segmentation-guided Re-ID experiments; therefore, it cannot be used as an example of automatic segmentation. Successes, failures, and partially successful cases are considered in this research instead of only positive ones.

In each case of data reuse, individual evaluation should be carried out. Data set registry provides information on the existing licensing conditions and access, however, all conditional permissions should be evaluated before redistribution or utilization. This thesis does not imply ethics approval and user field consent.

The validation data is used for the selection of the model, while the held-out test data is used only for evaluation. The external test data should never be used again for tuning after its performance was estimated. It is important to note if some data instances were not included because of the perceptual failure. The authors of the paper are the final proprietors of their work even though they utilize the AI technology to write the texts or program. However, the authors have to ensure that their project is correct, true, and corresponds to the university policies regarding the use of AI technology and the citation rules.

3.5 Standards and Technical Conventions

The study follows common technical practices. However, no claim is made about having an engineering certification unless it has been properly verified. The released score range and the known anatomical meaning of BCS are used to understand what the BCS labels represent [1], [2]. Labels for behavior use a versioned mapping, and Re-ID has a clear query-gallery protocol. That being said, these rules do make the evaluation more fair, but they do not officially certify the method.

The thesis is prepared according to the prescribed chapter format of BRAC University and bibliography style of IEEE. Dataset licensing, privacy requirements,

deployment safety and any separate IEEE-format submission requirement need to be reviewed independently. The use of IEEE reference format does not imply that the work has been submitted or approved for publication to the Journal.

3.6 Project Management Plan

The project is organized into several main stages: protocol verification, perception assessment, single-task reference models, cattle-centered alternatives, and multi-task comparison. Each stage provides the information needed for the next. Table 3.2 records the main project milestones. The timeline shows project progress, not scientific evidence.

Table 3.2: Research milestones and manuscript-integration stages. Recorded dates are retained from the existing evidence summary.^[E01,E02,E03,E23,E25,E28,E30]

Stage	Milestone	Record or deliverable
Preliminary work	Problem formulation and initial investigations	Research motivation
20 Sep 2026	ScienceDB grouping and SideView protocols	Recorded protocols
20–22 Sep	Perception and visual assessments	Feasibility findings
23 Sep	CVB and Beef protocol and RGB results	Recorded comparisons
23 Sep	RGB baselines and SideView data preparation	Completed single-task references
24 Sep	Cattle-centered task models	Completed task evaluations
MTL comparison	Hard-shared control, modular task-private model, and PCGrad optimization control	Completed held-out evaluations
Final synthesis	Integrated discussion, abstract, and conclusion	Completed manuscript synthesis

Computational resources were selected according to the needs of preprocessing, training, and evaluation. Small pipeline checks were done before full experiments, and reusable intermediate outputs reduced repeated preprocessing. Detailed hardware and execution records are kept for reproducibility instead of being included as part of the main scientific argument.

With versioned code, deterministic manifests, permanent data storage, and machine-readable results, each experiment is linked to its own set of settings. These records help make sure that the same steps are taken in all situations and give us the data we need to repeat the results that were published.

Contributions from authors, the order of authors, information about the committee, and submission details still need to be confirmed by the authors and supervisors. They are not guessed from what is happening in the repository.

3.7 Risk Management

There is a list of the main scientific and operational risks in Table 3.3. Some of these are related observations that cross train and test splits, identity claims that can’t be backed up, classes that are linked to the same data source, perception failures, wrong causal claims from combined model changes, and unfair comparisons caused by different training budgets.

Table 3.3: Scientific and operational risks; implemented controls do not eliminate all residual risks.^[E04,E05,E06,E07,E11,E24,E26,E29]

Risk	Consequence	Control / open issue
Correlated data	Optimistic evaluation	Burst/source/session grouping; identity may remain unknown.
Temporary IDs	False cross-cow claim	Use biological IDs only where verified.
Unavailable data	Blocked experiments	Approved SideView contingency; MultiCam still blocked.
Source/class link	Shortcut prediction	Source/class metrics; Walking is CVB-only.
Perception error	Missing/wrong target input	Track valid masks and omit failed samples without fabricating data.
Combined changes	Unsupported causal claim	Report performance for the whole setup rather than single parts.
Oracle masks	Overstated deployment readiness	Mark Run 6 as using ground-truth masks, not automatic tools.
Single-run results	Unknown training variability	Give single-run numbers without claiming statistical significance.
External mismatch	Misleading transfer	Lock label groupings prior to evaluation.
Limited resources	Incomplete comparisons	Run quick local checks, train in cloud, and postpone E2, E5, and stress tests.
Unclear license	Uncertain reuse rights	Check data licenses directly instead of assuming rights from web access.

The evaluation design also includes some task-level controls that are aimed at avoiding the mentioned problems. ScienceDB burst groups and Behavior videos/sessions remain in the corresponding data partitions. The identities of the biological cows in the SideView Protocol A that are used for representation learning are separated from the identities of evaluation. Matched samples are used in comparing RGB, perception-based BCS, and Behavior models. Re-ID is an oracle guided model when ground truth masks are used. Some limitations of the experiments have been identified. ScienceDB lacks confirmed cow identities, Walking has only one source, and

each task model is based on a single run. In the final MTL comparison, the task evaluation protocols should remain comparable, and any changes in the training budget should also be stated.

3.8 Economic Analysis

The records provided can only provide some data about the cost of research. They are not sufficient to perform cost analysis for deployment to farms. Different experiments were performed using different GPUs, preprocessing methods, storage, and billing system. It is difficult to compare the runtime of the experiment as cost because of these differences without having similar rates or bills. The table provided contains only the information that has been recorded.

Table 3.4: Selected recorded training/pipeline runtimes; these are not a total project or deployment budget.^[E08,E09,E14,E22,E24,E28,E29]

Item	Elapsed time	Cost evidence
BCS baseline	2,023.22 s	About USD 1.15 in research log; not invoice-reconciled.
Behavior baseline	821.47 s	Dollar costs were not tracked in the result file.
Re-ID RGB baseline	1,049.05 s training	No financial cost was saved in the artifact log.
BCS cattle-centered	550.80 s across recorded epochs	Vision preprocessing was run separately without an overall total.
Behavior cattle-centered	1,009.91 s training	Offline data caching took 3,492.3 seconds as a separate run.
Re-ID oracle GT-guided	2,350.67 s full pipeline	Covers the training time and Protocol A test evaluation.
Complete system	Not measured	Commercial payback and power usage were not measured.

In order to obtain cost estimates of deploying the model, data on the number and location of cameras, frequency of inference, hardware requirements, storage and maintenance is required. This kind of data is not currently present in the experimental benchmarks that the authors have used. Consequently, the authors can make no reliable cost savings, ROI or payback period estimates.

Chapter 4

Proposed Methodology

4.1 Design Process and Methodology Overview

The study tests different ways of showing the visual representation to the model and each task uses its own evaluation model. Each task has an RGB reference model and a cattle-centered version. These six single-task models establish task-specific performance baselines and serve as the empirical reference for evaluating three multi-task learning configurations: a monolithic hard-shared control architecture (E1), a modular task-private architecture (E3), and a gradient-projected hard-shared optimization control (E4).

The main design question is what information each task should receive. BCS mainly needs morphology, behavior recognition needs posture and temporal change, and Re-ID needs stable individual appearance. Because of this, the cattle-centered models all focus on the target cow, but they do not use exactly the same preprocessing. Figure 4.1 shows the overall comparison, and Table 4.1 lists the model configurations. Supporting records are indexed in Appendix A.

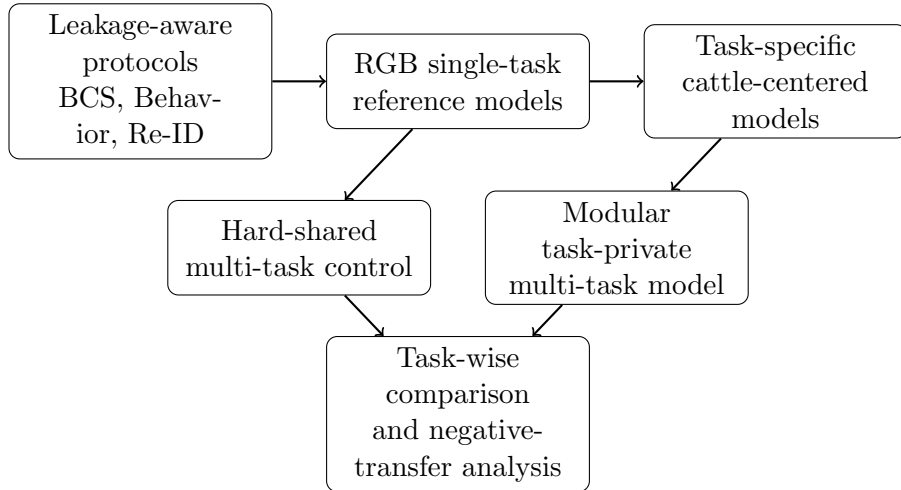


Figure 4.1: Comparative research framework illustrating single-task baselines, cattle-centered perception models, and unified multi-task evaluations across BCS, behavior recognition, and re-identification.

The perception enhanced models perform a number of modifications simultaneously. In BCS, the architecture modifies localization, cropping, and also uses mask infor-

Table 4.1: Configurations in the comparative research framework. Single-task reference models and unified multi-task architectures are evaluated across the matched held-out protocols for body condition scoring, behavior recognition, and individual re-identification.

Configuration	Role	Evaluation Protocol
RGB BCS baseline	Single-task reference	Full 8,040-image test.
Single-frame RGB behavior baseline	Single-task reference	Full 809-sequence test.
RGB Re-ID baseline	Single-task reference	Protocol A Barn and Snapshot retrieval.
Perception-enhanced BCS	Crop and binary-mask guidance	Matched 7,549-image evaluation.
Perception-enhanced temporal behavior	Crop, binary mask, and Conv1D aggregation	Matched 780-sequence evaluation.
Oracle segmentation-guided Re-ID	Ground-truth crop and mask guidance	Protocol A Barn and Snapshot retrieval.
Hard-shared multi-task control (E1)	Monolithic joint representation reference	Matched multi-task evaluation across all three tasks.
Modular task-private multi-task model (E3)	Task-private residual adapter architecture	Matched multi-task evaluation across all three tasks.
PCGrad hard-shared optimization control (E4)	Gradient-projected shared representation control	Matched multi-task evaluation across all three tasks.

mation. In behavior recognition, the architecture modifies visual features and the temporal aggregation strategy. In Re-ID, the architecture modifies crop geometry and introduces an oracle mask channel. Thus, these comparisons evaluate the performance of a complete model architecture, but they cannot provide us with the effect of each component independently. The backbone of reference RGB models is a common ImageNet pretrained ResNet-18 for all three tasks.

4.2 Preliminary Designs and Model Specifications

4.2.1 Representation Alternatives

However, for each task, a different prediction head is used: cumulative ordinal logits for BCS, categorical logits for behavior recognition, and an identity classifier with retrieval embedding for Re-ID. The task protocol remains the same to allow the more meaningful comparison between RGB and cattle-centered input.

We chose the localization and segmentation approaches based on the experiments with pretrained perception models. The success rate of detection and duration of the process were estimated during the research. Mask overlap was estimated in cases where ground-truth masks of SideView were available. The RT-DETR-L and subsequent SAM 2.1 served as a convenient automatic pipeline for ScienceDB and Behavior data. Masks that were made public were used independently in an oracle-guided Re-ID experiment. System diagram demonstrates the task-specific perception and input pipelines for three monitoring tasks.

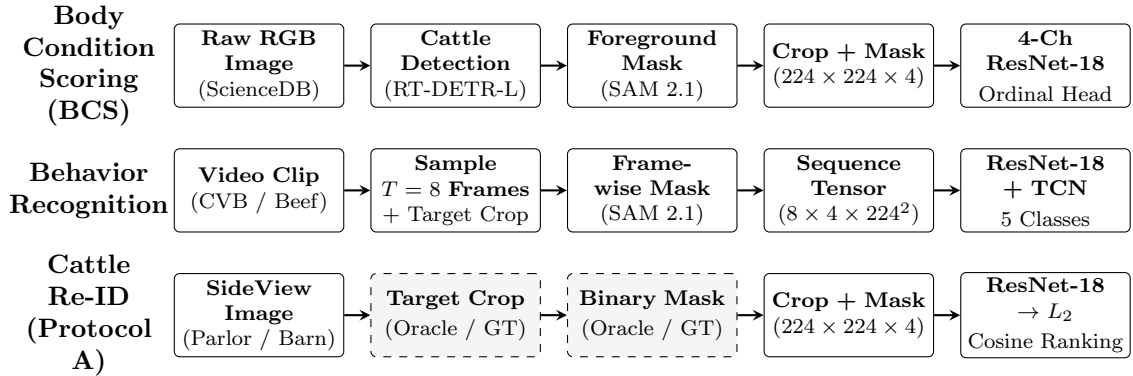


Figure 4.2: Task-specific cattle-centered input pipelines. Target cow localization and soft segmentation masks are generated automatically (RT-DETR-L and SAM 2.1) for BCS and behavior, and obtained from ground-truth annotations for re-identification.

4.2.2 Baseline RGB Body Condition Scoring

The BCS baseline uses an ImageNet-initialized ResNet-18 with four independent cumulative threshold logits for five ordered classes. For a class index $y \in \{0, 1, 2, 3, 4\}$, the target for threshold k is

$$t_k = \mathbb{I}(y > k), \quad k = 0, 1, 2, 3. \quad (4.1)$$

If $g_k = \mathbf{w}_k^\top \mathbf{h} + b_k$ is the corresponding logit, the batch loss is

$$L_{\text{BCS}} = -\frac{1}{4B} \sum_{i=1}^B \sum_{k=0}^3 [t_{ik} \log \sigma(g_{ik}) + (1 - t_{ik}) \log(1 - \sigma(g_{ik}))]. \quad (4.2)$$

The predicted class is found by counting how many threshold probabilities are above 0.5. It is then mapped back to the physical BCS scale:

$$\hat{y} = \sum_{k=0}^3 \mathbb{I}(\sigma(g_k) > 0.5), \quad \widehat{\text{BCS}} = 3.25 + 0.25\hat{y}. \quad (4.3)$$

The targets are cumulative, but the model uses independent ordinal binary cross-entropy. Therefore, the method is described as `ordinal_bce`, not CORAL, because CORAL uses a different rank-consistency method [19]. The model was trained for 30 epochs with batch size 64, AdamW, an initial learning rate of 10^{-4} , weight decay 10^{-4} , a cosine learning-rate schedule, and seed 42. The checkpoint with the lowest validation MAE was kept.

4.2.3 Perception-Enhanced Body Condition Scoring

The perception-enhanced BCS model first localizes and segments the cow in ScienceDB. RT-DETR-L detects cattle using the COCO cattle class with a confidence threshold of 0.25. If more than one cow is detected, the largest bounding box is selected, and confidence is used to break a tie. The selected box is expanded by 5% in each direction, clipped to the image boundary, and used for both the RGB crop and the SAM 2.1 prompt. If detection or segmentation fails, the image is recorded as a perception failure and excluded. No artificial replacement input is used.

The RGB crop and binary mask are resized together to 224×224 . The RGB channels use ImageNet normalization, while the mask stays as an unnormalized binary channel in $\{0, 1\}$. The first ResNet-18 convolution is changed from three input channels to four. The pretrained RGB weights are kept, and the new mask-channel weights are initialized using the mean of the RGB filters. The ordinal head, target encoding, loss, training schedule, and augmentation policy otherwise follow the RGB BCS baseline. This adds 3,136 trainable parameters, increasing the model from 11,178,564 to 11,181,700 parameters. The model uses a binary mask channel; it is not a calibrated soft-mask model.

4.2.4 Baseline Single-Frame RGB Behavior Recognition

Pose and viewpoint were not added to the first paired models since their utility and transferability for the tasks had not been proven. It should be noted that this does not exclude them from further separate ablation studies. Behavior uses the middle frame of each canonical sample. The ImageNet pretrained ResNet-18 is employed together with a five-class linear classifier. Five classes are Standing, Lying, Feeding, Drinking, and Walking. The model is trained using cross-entropy loss. For the CVB, the target tracklet information obtained from the dataset is used, and Kaggle Beef provides one cow for each clip.

Out of the 30 trained epochs, epoch 25 is selected as the checkpoint with respect to the Macro-F1 score in the validation. The baseline does not have any kind of

temporal aggregation intentionally; it seeks to see what is the extent of information that can be derived from just one RGB frame in a regular five-class scenario. This is why the performance metrics include overall accuracy, balanced accuracy, and class-specific results. The significance of this point is the fact that Walking is present only in the CVB dataset.

4.2.5 Perception-Enhanced Temporal Behavior Recognition

The temporal behavior model uses eight approximately evenly spaced frames from each labeled interval instead of only the midpoint frame. Its input tensor has shape $[B, 8, 4, 224, 224]$. Each time step contains an RGB crop and a binary mask. For CVB, the target-tracklet box at the sampled frame is used to prompt SAM 2.1, and the same box is used for the RGB and mask crops. The model does not replace the target with a nearby cow or with the full image. For the single-cow Beef clips, RT-DETR-L provides the main cow box when available, followed by a positive center prompt to SAM 2.1. If the detector finds no cow, an image-center prompt is used. A sequence is excluded if any required mask is empty or invalid.

The training and validation cache kept 4,271 of 4,465 candidate sequences: 3,641 for training and 630 for validation. It contains 34,168 RGB frames and 34,168 non-trivial binary masks. There were 194 exclusions because of empty-mask failures, and all 145 Walking candidates in training and validation were kept. The held-out test comparison separately kept 780 of 809 samples, including 422 CVB and 358 Beef sequences.

First, each frame is transformed using a four-channel ResNet-18, and hence a feature vector of 512 dimensions is obtained. The sequence H is $[B, 8, 512]$ in size and is first transposed and then used for the temporal convolution. For the first Temporal block, Conv1D with kernel size three is used to down project the channel sizes from 512 to 256. Following that, BatchNorm, GELU, Dropout 0.2, and one-width residual projection come up. In the second Conv1D block, the kernel size used is three and the feature size is of 256 channels. Normalization, activation, and dropout are applied along with identity residual. Adaptive average pooling along the time axis leads to one 256-dimensional feature vector which feeds into the linear classifier of five classes. The model being proposed here is a light residual Conv1D model which accepts sparsely sampled frame features but does not use optical flow, video transformer or deep dilated causal TCN.

4.2.6 Baseline RGB Cattle Re-Identification

The total number of trainable parameters in the model is 11,903,621. For RGB Re-ID, the SideView Protocol A is used in the baseline. To learn representation, 41 cows are selected from parlor only. 12,753 training and 2,683 validation images are used. The total evaluation is performed on 69 different cows. Gallery images are 36,811 of parlor. Barn query consists of 25,260 images of all the 69 cows. On the other hand, snapshot query consists of 607 images of 63 cows. The cows for representation learning and evaluation are distinct.

ResNet-18 produces a raw 512-dimensional feature $\mathbf{h}$. A 41-class linear classifier is trained on this feature with cross-entropy loss. Retrieval uses the L2-normalized

version:

$$\mathbf{z} = \frac{\mathbf{h}}{\|\mathbf{h}\|_2}, \quad \ell = W\mathbf{h} + \mathbf{b}. \quad (4.4)$$

During evaluation, cosine similarity is calculated as the dot product between normalized query and gallery embeddings. Gallery images are ranked for each query, and performance is reported using Rank- k and mean average precision (mAP). The model was trained for 30 epochs, and the validation-selected representation was evaluated once under the held-out Protocol A conditions.

4.2.7 Oracle Segmentation-Guided Re-Identification

For the segmentation-guided Re-ID model, since the ground-truth masks of the target cows are provided by SideView, the experiment is oracle-based. The non-zero area of each mask is used to generate a bounding box which is then padded to be slightly larger than the mask by about 5%. The same box coordinates are employed to crop the RGB image and the mask. They are then resized to 224x224. The RGB image is normalized with ImageNet values whereas the mask is binary having values of 0 and 1. The RGB image is not multiplied with the mask. The first convolution layer is modified to receive four input channels. The filter parameters for the newly added mask channel are initialized using the average of RGB filter parameters.

This comparison tests whether reliable target-centered cropping and explicit foreground information help cross-setting retrieval. It does not test an automatic segmentation pipeline, and it does not separate the effect of cropping from the effect of the fourth channel. The extra 3,136 weights increase the trainable parameter count from 11,197,545 to 11,200,681.

4.3 Data Collection and Preparation

4.3.1 Data Sources and Scientific Roles

All experiments use existing datasets; no new farm data were collected for this thesis. ScienceDB is the main BCS dataset, CVB and Kaggle Beef form the main behavior dataset, and SideViewCows2026 is the main Re-ID dataset. External datasets have separate stress-test roles and are not described as evaluated unless supporting results exist. Table 4.2 summarizes these roles.

The rest of the training and retrieval steps are identical to the RGB Re-ID baseline. ScienceDB has 53,566 RGB images and five BCS values varying between 3.25 and 4.25. The biological cow ID for ScienceDB metadata is not validated. Therefore, the 5,653 connected and fixed burst groups will be the most accurate grouping method. In terms of this type of evaluation, ScienceDB is known as burst group disjoint, passage disjoint, or sequence safe, but not unseen cow evaluation. In the Behavior manifest, there are 2,481 CVB segments and 2,793 Beef clips, having 267 protected groups and 5,274 samples.

For CVB samples, grouping is based on source video, while Beef samples' grouping is based on recording session or source video. Walking category is present in CVB. Consequently, Behavior evaluation is source/session group disjoint and not cow disjoint, and accuracy alone will not suffice for this evaluation. Preprocessing, spatial feature extraction, temporal sequence aggregation, and multitask learning are done

Table 4.2: Primary and external dataset roles. An assigned external role does not imply that evaluation has been executed.^[E04,E05,E06,E07]

Source	Role / scale	Boundary
ScienceDB	Primary BCS; 53,566 RGB images	5,653 burst groups; no verified biological cow IDs.
CVB + Beef	Primary Behavior; 5,274 sequences/clips	Source-video/session grouped; Walking is CVB-only.
SideViewCows2026	Primary Re-ID; 80,260 image/mask pairs; 110 cows	Protocol A learning/evaluation identities are disjoint.
Ruchay 2026 / Dryad	External BCS	Planned validation; score support and modality differ.
MmCows / CBVD-5	External Behavior	Planned stress tests; label/temporal compatibility required.
BECA-L / BECA-D	External Re-ID	Planned longitudinal and population-scale stress tests.
MultiCamCows2024 / OpenCows2020	Blocked / legacy	Neither is the active primary benchmark.

via PyTorch, Torchvision, OpenCV, and Ultralytics perception API, respectively. CSV manifests are deterministic and include all image samples, biological groups, and data split partitions.

SideView contains 80,260 RGB images, 80,260 matching binary masks, and 110 biological cow identities. Protocol A separates 41 cows used for representation learning from 69 cows used for evaluation. This supports an identity-disjoint interpretation for the Re-ID comparisons, unlike the BCS and behavior protocols.

4.3.2 Label Harmonization and Split Protection

ScienceDB keeps its original physical BCS values. CVB labels for resting-standing, resting-lying, grazing, drinking, and walking are mapped to the five common behavior classes. Beef contributes stand, lie, eat, and drink, while rumination is excluded. This label mapping is fixed before evaluation. SideView cow identities are used only in their assigned learning or evaluation role.

Table 4.3: Canonical BCS, Behavior, and Re-ID experimental units.^[E04,E05,E06]

Source	Partition	Examples	Groups
ScienceDB	Train	37,045	3,958 bursts
ScienceDB	Validation	8,481	850 bursts
ScienceDB	Test	8,040	845 bursts
CVB + Beef	Train	3,785	184 source/session
CVB + Beef	Validation	680	39 source/session
CVB + Beef	Test	809	44 source/session
SideView learning	Train	12,753	41 cows
SideView learning	Validation	2,683	41 cows
SideView Protocol A	Gallery (parlor)	36,811	69 cows
SideView Protocol A	Query (barn)	25,260	69 cows
SideView Protocol A	Query (snapshots)	607	63 cows

The splits protect the strongest available source of provenance. ScienceDB keeps each connected burst group inside one partition. All CVB segments from the same source video stay in one partition, and all Beef clips from the same recording session stay together. SideView Protocol A separates cow identities between learning and evaluation and also defines the acquisition setting of the gallery and queries. The implemented checks found no forbidden group or identity overlap. However, this does not prove that every possible form of leakage has been removed.

4.3.3 Perception Feasibility and Component Selection

Before choosing the perception pipeline, we compared several off-the-shelf components. The assessment separated output-return rate from actual accuracy and measured mask overlap only where ground truth was available. The results supported RT-DETR-L and SAM 2.1 as practical components. They also showed that getting a valid output from a perception model does not automatically mean that the output contains the anatomy needed by the downstream task.

Pose was not included in the first paired task configurations. A small manual review on ScienceDB showed unreliable results on rear-view images, and the available land-

Table 4.4: Feasibility measurements; detection and pose return rates are not accuracy.^[E11]

Candidate	Sample / measure	Result
YOLOv8s	300 images; any cow returned	223/300; median 24.5 ms
Faster R-CNN v2	Same detection measure	285/300; median 470.8 ms
RT-DETR-L	Same detection measure	283/300; median 94.3 ms
RT-DETR + SAM 2.1	100 SideView images; mask overlap	Mean IoU 0.9216; Dice 0.9530
Oracle box + SAM 2.1	Ground-truth-derived prompt	Mean IoU 0.9468; diagnostic
Pose backbones	300 images each; output return	243/300; not pose accuracy

mark set did not directly describe the intended BCS anatomy. A separate real-cattle viewpoint classifier reached 86.26% accuracy, 84.96% balanced accuracy, and 0.8573 Macro-F1 on its own held-out test of 131 images. Viewpoint was not included in the first paired task models because its downstream transfer had not been validated at that time. Camera identity was not used as a replacement for viewpoint. These statements describe the design choice for the original paired models; they do not mean that pose or viewpoint is permanently rejected.

4.4 Implementation of Selected Design

The checkpoint was picked up solely on the basis of results on validation sets, while test sets remained entirely out of training and model selection. All configuration parameters, losses per epoch, check points’ checksums, and evaluations rankings are stored in machine-readable manifests to enable scientific reproducibility.

4.4.1 Monolithic Hard-Shared Multi-Task Control Architecture

A monolithic hard-shared multi-task model (E1) is used as a baseline for parameter sharing between the three tasks. In this case, all three tasks share a single spatial feature extractor, i.e., four-channel ResNet-18, as their backbone.

The model accepts task-specific input tensors whose spatial dimensions are standardized to 224×224 pixels across four channels, comprising three ImageNet-normalized RGB channels and one unnormalized binary cow mask:

$$\mathbf{X}_{\text{BCS}} \in \mathbb{R}^{B_{\text{BCS}} \times 4 \times 224 \times 224}, \quad (4.5)$$

$$\mathbf{X}_{\text{Beh}} \in \mathbb{R}^{B_{\text{Beh}} \times 8 \times 4 \times 224 \times 224}, \quad (4.6)$$

$$\mathbf{X}_{\text{ReID}} \in \mathbb{R}^{B_{\text{ReID}} \times 4 \times 224 \times 224}. \quad (4.7)$$

The shared spatial backbone maps each four-channel frame to a 512-dimensional embedding $\mathbf{h} \in \mathbb{R}^{512}$. Each task then routes this representation to its task-specific prediction head:

1. **Body Condition Scoring:** Body Condition Scoring: The common feature $\mathbf{h}$ is passed to a cumulative ordinal linear head with 2,052 parameters. Four independent logits of thresholds for k values of 0, 1, 2, and 3 are produced by the BCS model. The training of the BCS model is conducted with the help of ordinal binary cross-entropy loss (L_{BCS}).
2. **Cattle Behavior Recognition:** Cattle Behavior Recognition: The same four-channel ResNet-18 spatial backbone is used to process each frame of 8 in the video sequence. The output frame features $\mathbf{H}$ have the shape of $[B_{\text{Beh}}, 8, 512]$. This sequence is further passed to the TCN head, which has 723,973 parameters. This includes two Conv1D residual blocks with a kernel width of three and 256 channels, together with BatchNorm, GELU, and dropout of 0.2. After that, a 256 dimensional feature is extracted by temporal adaptive average pooling. This feature is then passed to a five-class linear classifier. The Behavior model is trained using categorical cross-entropy loss (L_{Beh}).
3. **Individual Cattle Re-Identification:** Re-Identification of Individual Cattle: In the representation learning part, the standard 512 dimensional $\mathbf{h}$ is passed into the 41-class linear classification model with bias having the 41 identities used for the parlor training as the goal. The model has 21,033 trainable parameters and trained using cross-entropy loss (L_{ReID}). During the open-set retrieval evaluation on novel cattle, embeddings are L2 normalized to have a unit norm.

The monolithic hard-shared multi-task control contains exactly 11,926,706 trainable parameters.

4.4.2 Modular Task-Private Multi-Task Architecture

The monolithic hard-shared multi-task model consists of 11,926,706 trainable parameters exactly. In the modular task-private multi-task model (E3), we intend to see if an increase in the capacity of the model specific to each task helps in overcoming the negative transfer without changing the shared visual model architecture. The model is made identical in all aspects of the backbone architecture, task heads, inputs, datasets, batch size, loss function, optimizer, training budget, and model checkpoint except that there are three task-private residual adapters used in the model. This leads to an increase in the number of trainable parameters by 395,904 (3.32%). As such, the variations noted herein are the result of full modularization, which is not possible due to routing alone. Figure 4.3 illustrates the structural comparison between the monolithic hard-shared version against the modular task-private E3 version. The hard-shared version also forms the exact architecture used for E4. Also shown in the diagram is the bottleneck adapter design utilized by E3. The architectural modification consists of inserting one independent residual bottleneck adapter between the shared 512-dimensional representation $\mathbf{h}_{\text{shared}} \in \mathbb{R}^{512}$ and each respective task head. For task $t \in \{\text{BCS}, \text{Beh}, \text{ReID}\}$, the adapter consists of:

1. A linear down-projection from 512 to 128 dimensions with weights $\mathbf{W}_{\text{down}}^{(t)} \in \mathbb{R}^{128 \times 512}$ and bias $\mathbf{b}_{\text{down}}^{(t)} \in \mathbb{R}^{128}$;
2. Layer normalization over the 128-dimensional bottleneck;

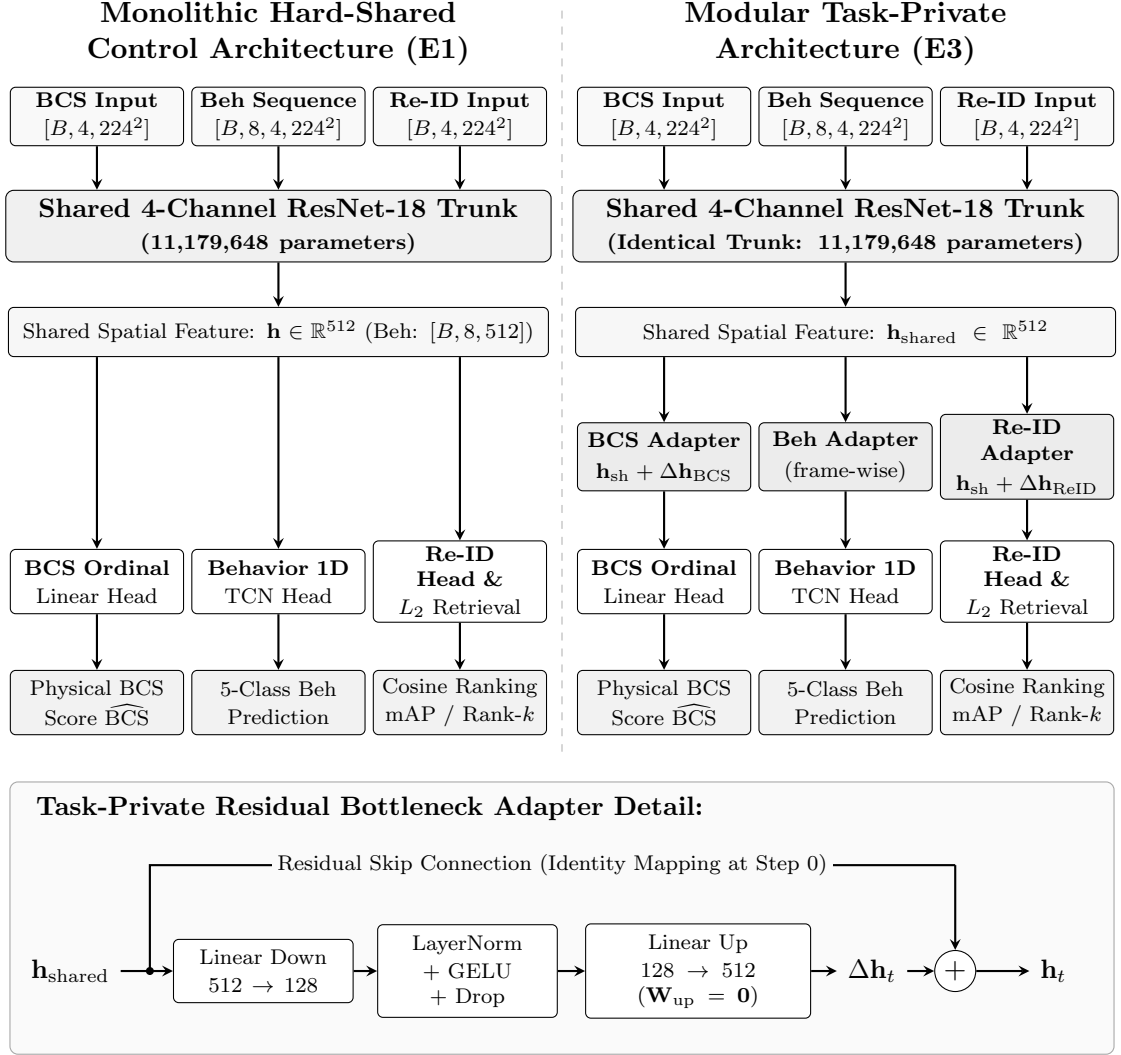


Figure 4.3: Architectural comparison between the monolithic hard-shared multi-task control (E1, left) and the modular task-private architecture (E3, right) with residual bottleneck adapters.

3. A GELU activation function followed by dropout ($p = 0.1$);
4. A linear up-projection from 128 back to 512 dimensions with weights $\mathbf{W}_{\text{up}}^{(t)} \in \mathbb{R}^{512 \times 128}$ and bias $\mathbf{b}_{\text{up}}^{(t)} \in \mathbb{R}^{512}$.

The adapter output $\Delta \mathbf{h}_t \in \mathbb{R}^{512}$ is added residually to the shared spatial representation to produce the task-adapted feature $\mathbf{h}_t$:

$$\mathbf{h}_t = \mathbf{h}_{\text{shared}} + \Delta \mathbf{h}_t. \quad (4.8)$$

In the temporal behavior model, this adapter operates frame-wise on each of the 8 temporal feature vectors before the adapted sequence $\mathbf{H}_{\text{Beh}} \in \mathbb{R}^{B_{\text{Beh}} \times 8 \times 512}$ is routed to the temporal convolutional network.

Each task-private adapter contains exactly 131,968 parameters, yielding 395,904 total private parameters across the three tasks (3.21% of the model’s total parameters). The modular multi-task architecture comprises 12,322,610 trainable parameters, representing an increase of 395,904 parameters (+3.32% relative capacity) over the hard-shared control.

Identity Initialization of Residual Adapters

The up-projection layer of each task adapter is initialized with zero weights and zero biases:

$$\mathbf{W}_{\text{up}}^{(t)} = \mathbf{0}, \quad \mathbf{b}_{\text{up}}^{(t)} = \mathbf{0}. \quad (4.9)$$

At step zero of training, this ensures that the adapter contribution is strictly zero:

$$\Delta \mathbf{h}_t = \mathbf{0} \implies \mathbf{h}_t = \mathbf{h}_{\text{shared}}. \quad (4.10)$$

This initialization does not imply inheriting representations from a pretrained hard-shared checkpoint; rather, it guarantees that at the start of joint training, each task-specific head receives the direct, unperturbed output of the modular model’s own shared backbone. Task-private feature specialization $\Delta \mathbf{h}_t$ then develops progressively as gradient updates accumulate during training.

4.4.3 Gradient-Projected Hard-Shared Multi-Task Optimization

In the third multi-task model, which is known as E4, the researchers evaluate whether the gradient conflict decrease during optimization process might lead to the decrease in negative transfer while keeping the architecture of the model unchanged. Thus, E4 does not acquire any extra architectural capabilities. E4 keeps the same parameter structure as the monolithic hard-shared model E1. There is one shared four-channel ResNet-18 spatial backbone (11,179,648 parameters) along with the same three task-specific prediction heads (747,058 parameters). The latter ones are the following: BCS ordinal head, Behavior 1D TCN, and Re-ID classification head. So, E4 has exactly 11,926,706 trainable parameters, without task-private adapters and without routing gates. All the training datasets, input formats, batch sizes, task losses, optimizer parameters, learning rate schedules, task loss weights, and checkpoint selection policies are kept strictly the same as E1.

The controlled experimental intervention in E4 resides exclusively in how task-specific gradients are combined on the shared backbone parameters θ_{shared} during each multi-task training super-step. When training multi-task networks with hard parameter sharing, backward passes from distinct task losses frequently produce gradient vectors that point in conflicting directions, such that the gradient update for one task increases the loss of another [23]. To address this, E4 applies Projecting Conflicting Gradients (PCGrad; Yu et al., 2020) to the shared backbone gradients. The controlled experimental intervention in E4 resides exclusively in how task-specific gradients are combined on the shared backbone parameters θ_{shared} during each multi-task training super-step. When training multi-task networks with hard parameter sharing, backward passes from distinct task losses frequently produce gradient vectors that point in conflicting directions, such that the gradient update for one task increases the loss of another [23]. To address this, E4 applies Projecting Conflicting Gradients (PCGrad; Yu et al., 2020) to the shared backbone gradients. Within each super-step, three backward passes first compute task-specific gradients on the shared backbone:

$$\mathbf{g}_{\text{BCS}} = \nabla_{\theta_{\text{shared}}} L_{\text{BCS}}, \quad \mathbf{g}_{\text{Beh}} = \nabla_{\theta_{\text{shared}}} L_{\text{Beh}}, \quad \mathbf{g}_{\text{ReID}} = \nabla_{\theta_{\text{shared}}} L_{\text{ReID}}. \quad (4.11)$$

The task gradient vectors are evaluated pairwise in a randomized order deterministically shuffled at each step using random seed 2026. For task i and comparison task j ($i, j \in \{\text{BCS}, \text{Beh}, \text{ReID}\}, i \neq j$), directional conflict is determined by the sign of their inner product. If

$$\mathbf{g}_i^\top \mathbf{g}_j < 0, \quad (4.12)$$

the two gradients point in opposing directions. The gradient vector $\mathbf{g}_i$ is then projected onto the normal plane of $\mathbf{g}_j$:

$$\mathbf{g}_i \leftarrow \mathbf{g}_i - \frac{\mathbf{g}_i^\top \mathbf{g}_j}{\|\mathbf{g}_j\|_2^2 + \epsilon} \mathbf{g}_j, \quad (4.13)$$

where $\epsilon = 10^{-8}$ is a small constant for numerical stability. In accordance with the PCGrad formulation [23], the comparison vector $\mathbf{g}_j$ is the original, unmodified task gradient to prevent order-dependent distortion of the reference direction. If $\mathbf{g}_i^\top \mathbf{g}_j \geq 0$, no projection is applied.

Once all pairwise projections are complete, the resulting modified shared gradients $\mathbf{g}'_{\text{BCS}}$, $\mathbf{g}'_{\text{Beh}}$, and $\mathbf{g}'_{\text{ReID}}$ are summed to form the unified update vector for the shared parameters:

$$\mathbf{g}_{\text{shared}} = \mathbf{g}'_{\text{BCS}} + \mathbf{g}'_{\text{Beh}} + \mathbf{g}'_{\text{ReID}}. \quad (4.14)$$

Gradients are added together instead of being averaged in order for the experiment to remain consistent with the baseline model (E1). In the baseline model, the gradients are combined based on a weight of 1 for each task. Consequently, any deviations from the training regime can be attributed to the gradient projection method and not the learning rate.

Each task gets its unique prediction head, which is not shared between tasks. That is why, their gradients will not interfere with one another. For this reason, PCGrad is only applied to the shared ResNet-18 backbone, while the gradients of task heads

remain untouched:

$$\mathbf{g}_{\text{head}}^{(\text{BCS})} = \nabla_{\boldsymbol{\theta}_{\text{head}}^{(\text{BCS})}} L_{\text{BCS}}, \quad (4.15)$$

$$\mathbf{g}_{\text{head}}^{(\text{Beh})} = \nabla_{\boldsymbol{\theta}_{\text{head}}^{(\text{Beh})}} L_{\text{Beh}}, \quad (4.16)$$

$$\mathbf{g}_{\text{head}}^{(\text{ReID})} = \nabla_{\boldsymbol{\theta}_{\text{head}}^{(\text{ReID})}} L_{\text{ReID}}. \quad (4.17)$$

To monitor optimization dynamics throughout training without biasing model selection, the E4 training loop computes diagnostic metrics before and after projection at every super-step. For each task pair (i, j) , these include the unprojected inner product $\mathbf{g}_i^\top \mathbf{g}_j$, the unprojected cosine similarity $\cos(\mathbf{g}_i, \mathbf{g}_j) = \frac{\mathbf{g}_i^\top \mathbf{g}_j}{\|\mathbf{g}_i\|_2 \|\mathbf{g}_j\|_2}$, the binary conflict indicator $\mathbb{I}(\mathbf{g}_i^\top \mathbf{g}_j < 0)$, the gradient norms before and after projection, the total number of conflict projections triggered per super-step, and the post-projection cosine similarities. These diagnostics permit direct measurement of shared-backbone gradient-direction conflict during E4 training. Table 4.5 summarizes the structural components and exact parameter counts across all three multi-task configurations.

4.4.4 Multi-Task Joint Optimization Protocol

Because cattle datasets from different farms and recording modalities do not contain overlapping annotations across body condition, behavior sequences, and individual identities, joint multi-task training cannot employ single multi-label batches. Instead, an interleaved mini-batch schedule is executed.

Each optimization step is structured as a multi-task super-step comprising three sequential forward and backward passes:

1. **BCS mini-batch:** A batch of $B_{\text{BCS}} = 64$ samples is passed through the model. The ordinal binary cross-entropy loss L_{BCS} is computed, and gradients are backpropagated into the active parameters.
2. **Behavior mini-batch:** A batch of $B_{\text{Beh}} = 8$ video sequences (64 total frames) is passed through the model. The categorical cross-entropy loss L_{Beh} is computed, and gradients are backpropagated.
3. **Re-ID mini-batch:** A batch of $B_{\text{ReID}} = 32$ images is passed through the model. The identity cross-entropy loss L_{ReID} is computed, and gradients are backpropagated.
4. **Parameter update:** In the monolithic control (E1) and modular architecture (E3), shared backbone gradients are accumulated directly by standard summation ($\mathbf{g}_{\text{shared}} = \sum_t \nabla_{\boldsymbol{\theta}_{\text{shared}}} L_t$) before a single AdamW parameter update. In the PCGrad optimization control (E4), task gradients on the shared backbone are captured independently, pairwise conflicts are projected via PCGrad, and the projected gradients are summed ($\mathbf{g}_{\text{shared}} = \sum_t \mathbf{g}'_t$) before a single AdamW parameter update, with task-head gradients remaining unmodified.

Equal task weighting is enforced throughout training:

$$w_{\text{BCS}} = w_{\text{Behavior}} = w_{\text{ReID}} = 1.0. \quad (4.18)$$

Table 4.5: Architectural specification and parameter allocation across the monolithic hard-shared control (E1), modular task-private multi-task architecture (E3), and PCGrad hard-shared optimization control (E4). All three models share an identical four-channel ResNet-18 spatial feature extractor and identical task prediction heads. E3 introduces three independent residual bottleneck adapters (+3.32% capacity), whereas E4 preserves the exact E1 parameterization (11,926,706 parameters) and intervenes only in shared-backbone gradient projection.

Component / Specification	Hard-Shared Control (E1)	Modular Task-Private (E3)	PCGrad Hard-Shared (E4)
Input Channels	4 (RGB + Binary Mask)	4 (RGB + Binary Mask)	4 (RGB + Binary Mask)
Shared Spatial Backbone	4-channel ResNet-18	4-channel ResNet-18	4-channel ResNet-18
Shared Feature Dimension	512	512	512
BCS Prediction Head	Cumulative ordinal linear	Cumulative ordinal linear	Cumulative ordinal linear
Behavior Prediction Head	2-block residual 1D TCN	2-block residual 1D TCN	2-block residual 1D TCN
Re-ID Prediction Head	41-class linear classifier	41-class linear classifier	41-class linear classifier
Task-Private Adapters	None (direct routing)	3 residual adapters	None (direct routing)
Adapter Bottleneck Dim.	N/A	512 $\rightarrow$ 128 $\rightarrow$ 512	N/A
Optimization Distinction	Standard gradient sum	Standard gradient sum	PCGrad projection on shared gradients
Shared Backbone Parameters	11,179,648 (93.74%)	11,179,648 (90.72%)	11,179,648 (93.74%)
Task-Private Adapter Parameters	0 (0.00%)	395,904 (3.21%)	0 (0.00%)
BCS Adapter	–	131,968	–
Behavior Adapter	–	131,968	–
Re-ID Adapter	–	131,968	–
Task Prediction Head Parameters	747,058 (6.26%)	747,058 (6.06%)	747,058 (6.26%)
BCS Ordinal Head	2,052	2,052	2,052
Behavior 1D TCN Head	723,973	723,973	723,973
Re-ID Classification Head	21,033	21,033	21,033
Total Trainable Parameters	11,926,706	12,322,610	11,926,706
Capacity Difference vs E1	Baseline	+395,904 (+3.32%)	0 (0.00%)

Training runs for 30 epochs with 538 super-steps per epoch (16,140 total super-steps). This schedule is defined by the complete ScienceDB BCS training population ($N = 34,369$ images; batch size $B_{\text{BCS}} = 64$), where $\lceil 34,369/64 \rceil = 538$ batches per epoch establish the super-step cadence across all three tasks. Optimization uses AdamW with an initial learning rate of $\eta_0 = 10^{-4}$, weight decay of 10^{-4} , and a cosine annealing learning-rate schedule decaying to $\eta_{\min} = 10^{-6}$.

Model checkpoint selection is determined strictly by minimizing the predefined mean validation loss objective:

$$L_{\text{MTL},\text{val}} = \frac{1}{3} (L_{\text{BCS},\text{val}} + L_{\text{Behavior},\text{val}} + L_{\text{ReID},\text{val}}). \quad (4.19)$$

Held-out test sets were excluded from training, hyperparameter tuning, threshold calibration, and checkpoint selection according to the implemented evaluation protocol. Table 4.6 outlines the per-task batch parameters and joint optimization hyperparameters.

Table 4.6: Multi-task training protocol and joint optimization hyperparameters. Training executes interleaved task-specific mini-batches per super-step with gradient accumulation before parameter updates.

Target Task	Training Unit	Batch Size	Task Head	Loss Formulation
Body Condition Scoring	Single image	64	Ordinal threshold linear	Ordinal binary cross-entropy
Behavior Recognition	8-frame sequence	8 seqs	Residual 1D TCN	Categorical cross-entropy
Cattle Re-Identification	Single image	32	41-class classifier	Categorical cross-entropy
Shared Multi-Task Optimization Hyperparameters				
Training Budget	30 epochs with 538 super-steps per epoch (16,140 total super-steps; $\lceil 34,369/64 \rceil = 538$)			
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$)			
Learning Rate Schedule	Initial $\eta_0 = 10^{-4}$, cosine annealing decaying to minimum $\eta_{\min} = 10^{-6}$			
Task Loss Weighting	Equal weighting: $w_{\text{BCS}} = 1.0, w_{\text{Beh}} = 1.0, w_{\text{ReID}} = 1.0$			
Checkpoint Selection Rule	Minimum mean validation loss: $L_{\text{MTL},\text{val}} = \frac{1}{3}(L_{\text{BCS},\text{val}} + L_{\text{Beh},\text{val}} + L_{\text{ReID},\text{val}})$			
Shared Gradient Integration	E1/E3: standard accumulated shared gradients; E4: pairwise PCGrad-projected shared gradients, then summed			

4.4.5 Architectural Gradient Isolation and Claim Boundaries

The mathematical distinction in gradient routing across the three executed multi-task configurations is formal and structural:

- In the **monolithic hard-shared control (E1)**, the parameter vector of the shared backbone θ_{shared} directly receives gradient contributions from all three task loss functions:

$$\mathbf{g}_{\text{shared}} = \nabla_{\theta_{\text{shared}}} L_{\text{BCS}} + \nabla_{\theta_{\text{shared}}} L_{\text{Beh}} + \nabla_{\theta_{\text{shared}}} L_{\text{ReID}}. \quad (4.20)$$

Each task head $\theta_{\text{head}}^{(t)}$ receives gradients solely from its own loss L_t .

- In the **modular task-private model (E3)**, the shared backbone θ_{shared} intentionally receives accumulated gradients from all three tasks through the residual connection:

$$\mathbf{g}_{\text{shared}} = \sum_{t \in \{\text{BCS}, \text{Beh}, \text{ReID}\}} \nabla_{\theta_{\text{shared}}} L_t. \quad (4.21)$$

In contrast, each task-private adapter $\theta_{\text{adapter}}^{(t)}$ and task head $\theta_{\text{head}}^{(t)}$ receive gradient signals strictly from their corresponding task loss L_t , with $\nabla_{\theta_{\text{adapter}}^{(t)}} L_{t'} = \mathbf{0}$ for all $t' \neq t$.

- In the **gradient-projected hard-shared control (E4)**, the architecture is identical to E1, but the shared backbone parameter vector θ_{shared} receives the sum of pairwise conflict-projected task gradients:

$$\mathbf{g}_{\text{shared}} = \mathbf{g}'_{\text{BCS}} + \mathbf{g}'_{\text{Beh}} + \mathbf{g}'_{\text{ReID}}, \quad (4.22)$$

while each task head $\theta_{\text{head}}^{(t)}$ receives its unmodified, unprojected task-specific gradient $\nabla_{\theta_{\text{head}}^{(t)}} L_t$.

The gradient separation originates from the model architecture and training procedure. The gradient of one task head is isolated from the other to prevent its direct impact on the private adapter of another task. In E4, the conflicting part of the shared-backbone gradients is projected. However, such a technique neither theoretically guarantees that there would be no gradient conflict in the shared backbone nor ensures that there would be no negative transfer. The actual impact of the above-discussed routing and optimization techniques on the performance of generalization across the three tasks is analyzed through experiments in Chapter 5. In the system architecture, the process of multi-task super-step, gradient combination, and their separation is illustrated.

4.4.6 Evaluation Metrics and Protocol Harmonization

The BCS is assessed with MAE of physical score, Acc@0 (exact accuracy), Acc@1 (accuracy of predicting one neighboring class or up to ± 0.25 BCS units), balanced accuracy, and Macro-F1. For N samples,

$$\text{MAE}_{\text{BCS}} = \frac{1}{N} \sum_{i=1}^N |s_i - \hat{s}_i|. \quad (4.23)$$

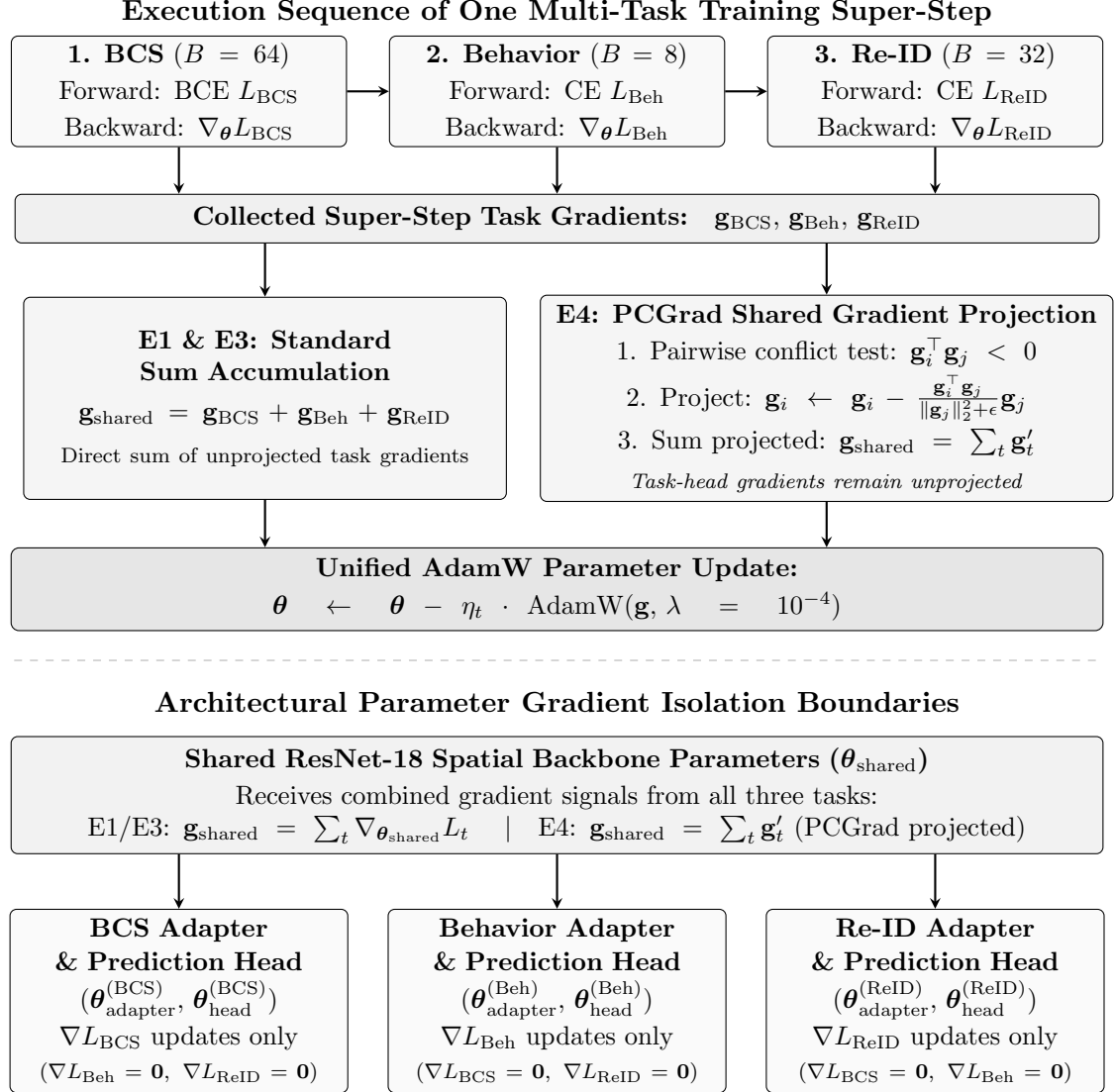


Figure 4.4: Multi-task training super-step protocol and architectural gradient routing. Tasks execute sequentially per super-step, with PCGrad (E4) projecting pairwise conflicting gradients on the shared backbone prior to parameter updates.

The behavioral analysis is conducted with overall accuracy, balanced accuracy, Macro-F1, per-class precision, recall, and F1, besides other source-specific scores. Balanced accuracy refers to the mean value of recall for the five classes; thus, it is useful for imbalanced data sets.

The metric to evaluate the performance of Re-ID system is known as Rank-k that determines the percentage of queries for which at least one correct identity is found in the first k gallery search results. Average Precision determines the quality of rankings attained by each query individually, whereas the mAP determines the average of all these Average Precision values. Barn-to-Parlor and Snapshots-to-Parlor are evaluated using the same parlor gallery of 36,811 images.

In order to maintain comparability, E1, E3, and E4 are evaluated using the same test population as the benchmark systems. The test datasets contain 7,549 images from ScienceDB test images for BCS, 780 retained video sequences for Behavior Recognition, and 69 unseen cattle identities for SideView Protocol A. The test population for Re-ID contains 25,260 barn queries and 607 snapshot queries.

Chapter 5

Result Analysis

5.1 Performance Evaluation

The experiments will consist of all the results of the single task and multiple tasks experiments, which consist of Hard-Shared Baseline, Modular Task-Specific Model, and PCGrad control experiment.

5.1.1 Body Condition Scoring Results

The Baseline RGB Run 1 serves as the principal reference result for the ScienceDB images used for testing which number 8,040 in total. The physical MAE achieved by the model is equal to 0.1848 BCS units, while the Acc@1 value is 86.74%. Most of the time, the prediction error was within ± 0.25 units, while predicting the score precisely was not easy. This should be considered as one estimate since the split was disjoint only among the burst groups and not cows.^[E08]

The Perception-Enhanced model (Run 4) requires both valid bounding-box detection and non-empty segmentation. The complete detector-segmenter pipeline succeeded on 7,549 of 8,040 test images, achieving 93.89% perception coverage (489 detection misses and two empty masks). To prevent sample attrition from confounding comparisons, the baseline model was re-evaluated on this identical 7,549-image matched subset.^[E24]

Table 5.1: ScienceDB BCS results. The valid direct comparison is between the two 7,549-image matched rows; the full Run 1 result is included as the canonical reference.^[E08,E24]

Configuration	N	MAE	Acc@1	Acc@0	Bal. acc.	Macro-F1
Run 1 RGB, full test	8,040	0.1848	86.74%	41.63%	40.41%	0.4110
Run 1 RGB, matched	7,549	0.1929	84.95%	40.84%	40.23%	0.4074
Run 4 cattle-centered	7,549	0.1709	89.40%	43.57%	39.70%	0.4039

The single-task experiment will begin by investigating whether the cattle-specific representation is superior to the existing RGB representation. On the other hand, the multiple task experiments will address the question of joint learning and ask whether negative transfer happens when the only feature representation is the same spatial representation. The multiple task experiments will also examine whether

the use of multiple task-specific models will compensate for the loss in performance, as well as whether PCGrad resolves the gradient conflict in the shared backbone network.

In the matched subset, the model with improved perception showed better results in primary ordinal metrics (table of summary), by decreasing the physical MAE from 0.1929 to 0.1709 BCS units and tolerance accuracy (Acc@1) from 84.95% to 89.40%. In contrast to this, there were no improvements for class-balanced metrics: the metrics such as balanced accuracy and Macro-F1 remained unchanged or even slightly worsened. Thus, cow-centered inputs have narrowed the gap between predicted and real intervals of scores but have not positively impacted class-balanced metrics.

5.1.2 Behavior Recognition Results

Our findings justify the use of localized crops and masks for assisting image understanding. It is impossible to determine the contribution of segmentation with a limited spatial scale since all three novel approaches have been implemented simultaneously. Additionally, our results do not include 6.11% of test cases where automatic perception did not work.

Within 809 test images according to the standard, the Baseline Single-Frame RGB model (Run 2) achieved 88.88% accuracy, and balanced accuracy was 71.72%, due to high class imbalance in the data. Poses in a static position, namely Lying, have been identified accurately, while an underrepresented class, such as Walking, has been recognized very poorly—only 5 out of 26 sequences were identified correctly (F1 = 0.2174).evidenceE09. As seen from the figure, the initial gap is high recall of feeding and resting postures while low recall of the dynamic and minority movements.

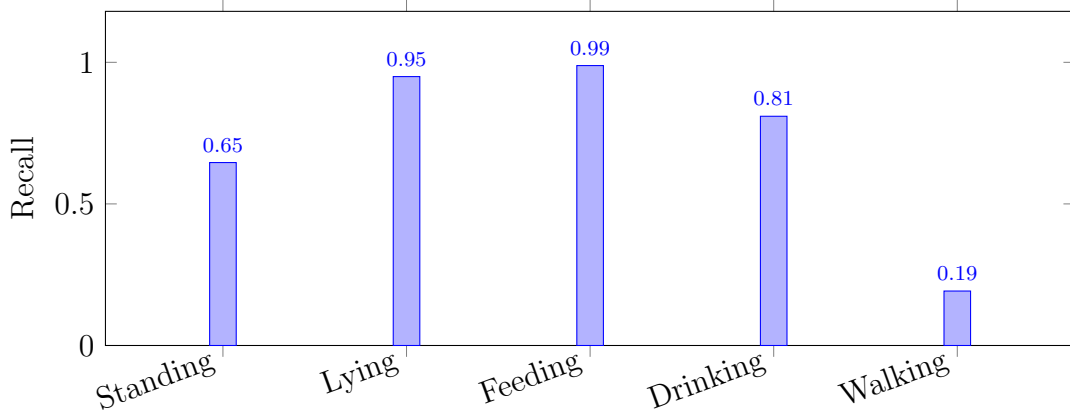


Figure 5.1: Run 2 full-test recall by class. Walking has 26 test observations and occurs only in CVB.^[E09,E12]

The perception pipeline retained 780 sequences (approximately 96.4% coverage), excluding 29 sequences due to pen stanchion occlusions. Re-evaluating Run 2 on this matched cohort establishes a fair baseline for the Perception-Enhanced Temporal model (Run 5).

As per Table 5.2, temporal foreground modeling has helped enhance balanced accuracy and test loss. While balanced accuracy went up from 71.30% to 74.43%, test loss has gone down from 0.5505 to 0.4430. However, overall accuracy was affected

Table 5.2: Behavior results. Direct comparison uses the same 780 retained test sequences; the 809-sample Run 2 row is the canonical full-test reference.^[E09,E27]

Configuration	N	Accuracy	Bal. acc.	Macro-F1	Loss
Run 2 RGB, full test	809	88.88%	71.72%	0.7413	0.5312
Run 2 RGB, matched	780	88.46%	71.30%	0.7378	0.5505
Run 5 perception + Conv1D	780	87.44%	74.43%	0.7397	0.4430

adversely and went down from 88.46% to 87.44%. This means that there is an improvement of the model for some classes but not necessarily in all measures.^[E27]

Table 5.3: Class-wise F1 on the matched 780-sequence Behavior test. Walking occurs only in CVB.^[E27]

Class	Run 2 RGB	Run 5 perception + Conv1D	Change
Standing	0.7556	0.7215	−0.0341
Lying	0.9512	0.9169	−0.0343
Feeding	0.9218	0.9465	+0.0247
Drinking	0.8430	0.8682	+0.0252
Walking	0.2174	0.2456	+0.0282

This is illustrated in the table of results where the dynamic and minority classes improved (Walking F1 went from 0.2174 to 0.2456) while the standing and lying posture classes slightly declined. The use of a foreground-guided and time-oriented approach improved some of the dynamic and minority classes but deteriorated some of the posture classes.

Table 5.4: Source-specific metrics on the matched Behavior test. Both sources occur in training; these are not external-transfer results.^[E27]

Source	Configuration	N	Accuracy	Bal. acc.	Macro-F1
CVB	Run 2 RGB	422	84.12%	60.62%	0.6541
CVB	Run 5 perception + Conv1D	422	80.09%	61.05%	0.6188
Beef	Run 2 RGB	358	93.58%	90.48%	0.9079
Beef	Run 5 perception + Conv1D	358	96.09%	94.26%	0.9414

Source-disjoint evaluation demonstrates pronounced domain dependence (Table 5.4). Performance on pasture cattle clips (Kaggle Beef) advanced to 96.09% accuracy and 0.9414 Macro-F1. Conversely, CVB surveillance saw overall accuracy decline from 84.12% to 80.09%. Because Run 5 simultaneously integrated foreground masks and 1D temporal convolution, these shifts cannot be attributed independently to mask guidance versus sequence modeling.

5.1.3 Cattle Re-Identification Results

The open set biometric identification experiment was done by using SideView Protocol A. The system scanned through a fixed database of 36,811 images from the

parlor using barn CCTV and handheld camera images. The experiment involved 69 cows that were not among the 41 training cows.^[E22,E29]

Table 5.5: Protocol A Barn-to-Parlor retrieval on 25,260 queries, 36,811 gallery images, and 69 held-out cows.^[E22,E29]

Configuration	Rank-1	Rank-5	Rank-10	mAP
Run 3 RGB	58.64%	78.19%	83.72%	38.32%
Run 6 oracle GT-guided	63.90%	77.10%	82.58%	40.68%
Change	+5.26 pp	−1.09 pp	−1.14 pp	+2.36 pp

In case of Barn-to-Parlor experiments, the Oracle Segmentation-Guided model (Run 6) enhanced the main retrieval performance in Table 5.5. Accuracy for Rank-1 increased from 58.64% to 63.90%, whereas the mAP increased from 38.32% to 40.68%. It indicates that the model performed better when identifying the right cow among the retrieved images. But the performance on the deeper ranks was more or less similar or even worse.

Table 5.6: Protocol A Snapshot-to-Parlor retrieval on 607 queries from 63 held-out cows and a 36,811-image gallery.^[E22,E29]

Configuration	Rank-1	Rank-5	Rank-10	mAP
Run 3 RGB	38.88%	57.17%	64.58%	27.05%
Run 6 oracle GT-guided	62.93%	75.29%	81.05%	40.42%
Change	+24.05 pp	+18.12 pp	+16.47 pp	+13.37 pp

The handheld Snapshot-to-Parlor evaluation yielded a substantially larger effect (Table 5.6). Under unconstrained mobile queries, Rank-1 accuracy rose from 38.88% to 62.93% (+24.05 pp), alongside an mAP rise from 27.05% to 40.42%. The oracle target-centered configuration produced its largest improvement under the Snapshot-to-Parlor setting, where acquisition differences were greater.

These results need to be considered in the context of oracle evaluation. Given that the ground truth masks have been used in Run 6 for cropping and channel concatenation, these results cannot confirm that the automation of the upstream segmentation task would yield the same improvements in practice. In addition, both target framing and mask guidance were combined.

5.1.4 Perception Coverage and Supporting Viewpoint Evidence

As is evident from the name itself, automatic filtering refers to keeping only the successful perception systems’ samples for analysis. The summary table provides information regarding how much data was retained in the evaluation.

For BCS, the percentage of data retained was 93.89% of the original dataset. In case of the Behavior dataset, 96.42% of sequences were retained, but 29 sequences were filtered due to the stanchion occlusion. All 80,260 ground-truth masks for SideView were available, which implies that SideView was used as an oracle baseline rather than being used as a completely automatic setting.

Table 5.7: Coverage of the cattle-centered inputs used in Runs 4–6. SideView coverage uses released ground-truth masks and is not an automatic-perception result. ^[E24,E26,E27,E29]

Evaluation	Retained / candidate	Coverage	Exclusions or boundary
Run 4 BCS test	7,549 / 8,040	93.89%	489 detector and 2 SAM failures.
Run 5 Behavior Train+Val	4,271 / 4,465	95.65%	194 invalid-mask exclusions.
Run 5 Behavior test	780 / 809	96.42%	29 perception exclusions.
Run 6 SideView Protocol A	All gallery/query images	100%	Paired masks supplied by the dataset.

A three-class viewpoint classifier for Front, Side, and Rear classes got an accuracy of 86.26% and Macro-F1 of 0.8573 on 131 images. However Viewpoint model was excluded from the multi-task pipeline ^[E31] because cross-domain feature transfer was not validated.

5.2 Multi-Task Learning Results

In order to assess the performance of all three tasks jointly, all models using multiple tasks were validated using the same matching held-out datasets as the single task models. For Body Condition Scoring, 7,549 ScienceDB images were employed, using burst-group-disjoint split, where identical passages were kept separate, but it was not a biological cow-disjoint split. For Behavior Recognition, 780 sequence samples were employed, each having 8 frames. In total, 6,240 frames were used for evaluation, consisting of 422 surveillance sequences from CVB, 358 feedlot sequences from Kaggle Beef dataset, while the Walking class was only present in the CVB dataset. For Cattle Re-Identification, SideView Protocol A was employed using 69 unseen cattle and a gallery containing 36,811 parlor images, together with 25,260 barn queries and 607 snapshot queries, using the provided oracle target masks. Only the best checkpoint was determined based on the lowest combined validation loss. The test sets were never seen during training and model selection and only evaluated once after training was done.

5.2.1 Body Condition Scoring Multi-Task Outcomes

The Table 5.8 demonstrates that BCS performance deteriorated when hard-shared multi-task approach (E1) was applied relative to the performance of the baseline (Run 4). Physical mean absolute error became equal to 0.1709 for Run 4 and 0.1788 for E1. Other metrics like tolerance accuracy and balanced performance metric became worse as well. Ordinal cross-entropy loss became somewhat better but did not contribute to the improvement of BCS score prediction.

Neither modular adapters nor gradient projection overcame this performance deficit. Introducing task-private bottleneck adapters (E3) degraded performance further,

Table 5.8: Multi-task Body Condition Scoring results on the matched held-out ScienceDB test population ($N = 7,549$ images). For MAE and Loss, lower values are better ($\downarrow$); for Acc@1, Acc@0, Balanced Accuracy, and Macro-F1, higher values are better ($\uparrow$). All models share the four-channel ResNet-18 spatial trunk.^[E24,E36,E37,E38]

Configuration	MAE	Acc@1	Acc@0	Bal. Acc.	Macro-F1	Loss
Single-Task Reference (Run 4)	0.1709	89.40%	43.57%	39.70%	0.4039	0.4403
Monolithic Hard-Shared (E1)	0.1788	88.44%	41.10%	35.70%	0.3605	0.4175
Modular Task-Private (E3)	0.1916	86.32%	38.47%	33.13%	0.3291	0.4555
PCGrad Hard-Shared (E4)	0.1828	87.84%	40.23%	34.97%	0.3516	0.4181

resulting in the highest error across all models (MAE = 0.1916). For E4, PCGrad with hard sharing managed to recoup some of the lost performance. In the process, the MAE was reduced to 0.1828 and the tolerance accuracy improved compared to E3. However, E4 was unable to match the performance of the single-task model where the MAE was 0.1709. All the three multitask models performed worse than the single-task model in the key physical and balanced metrics. This means that there was negative transfer for BCS in multitasking.

5.2.2 Behavior Recognition Multi-Task Outcomes

Joint training under naive hard sharing (E1) produced severe degradation across balanced evaluation measures (Table 5.9). Balanced accuracy plunged from 74.43% in the single-task reference to 67.30%, accompanied by sharp declines in overall accuracy and Macro-F1. The minority Walking class was disproportionately impacted, decreasing from an F1 of 0.2456 to 0.0408. Modular adapters (E3) offered only selective, loss-specific relief; while E3 improved test loss and overall accuracy, balanced accuracy eroded further (66.70%) and Walking F1 fell to 0.0000.

Table 5.9: Multi-task Behavior Recognition results on the matched held-out test population ($N = 780$ sequences, $T = 8$ frames). Higher values are better ($\uparrow$) for accuracy and F1 metrics; lower values are better ($\downarrow$) for loss. The single-task reference is Run 5.^[E27,E36,E37,E38]

Configuration	Accuracy	Bal. Acc.	Macro-F1	Loss	Walking F1
Single-Task Reference (Run 5)	87.44%	74.43%	0.7397	0.4430	0.2456
Monolithic Hard-Shared (E1)	85.00%	67.30%	0.6866	0.6226	0.0408
Modular Task-Private (E3)	85.77%	66.70%	0.6755	0.4193	0.0000
PCGrad Hard-Shared (E4)	86.92%	69.15%	0.7114	0.5454	0.0909

The PCGrad optimization control (E4) was the most effective multi-task variant for Behavior, partially mitigating hard-sharing degradation. E4 raised balanced accuracy to 69.15% and recovered Walking F1 to 0.0909 (more than doubling E1). Moreover, on CVB surveillance (Table 5.10), E4 reached an overall accuracy of 81.04%, marginally surpassing the single-task reference. However, source-disjoint results emphasize that overall accuracy and class-balanced fidelity diverge: CVB balanced accuracy (55.30%) and Walking F1 under E4 remained far below the dedicated single-task reference (61.05% and 0.2456, respectively). Thus, while PCGrad

provided clear optimization-level mitigation, it did not fully restore single-task discriminative power on difficult minority states.

Table 5.10: Multi-task Behavior Recognition performance breakdown across acquisition sources: CVB surveillance ($N = 422$) and feedlot camera clips (Kaggle Beef, $N = 358$). Note that the minority Walking class is exclusive to CVB.^[E27,E36,E37,E38]

Configuration	CVB Surveillance ($N = 422$)			Kaggle Beef Feedlot ($N = 358$)		
	Accuracy	Bal. Acc.	Macro-F1	Accuracy	Bal. Acc.	Macro-F1
Single-Task (Run 5)	80.09%	61.05%	0.6188	96.09%	94.26%	0.9414
Hard-Shared Control (E1)	76.78%	48.90%	0.5402	94.69%	93.11%	0.9236
Modular Task-Private (E3)	80.33%	60.61%	0.6026	92.18%	87.29%	0.8849
PCGrad Hard-Shared (E4)	81.04%	55.30%	0.6141	93.85%	92.05%	0.9158

5.2.3 Cattle Re-Identification Multi-Task Outcomes

The dedicated single-task oracle model (Run 6) maintained the strongest performance across both query settings, achieving a Rank-1 accuracy of 63.90% on Barn queries and 62.93% on Snapshot queries, with mAP exceeding 40% in both environments (Table 5.11). Monolithic hard sharing (E1) degraded these principal retrieval measures considerably, dropping Barn Rank-1 to 57.38% and Snapshot Rank-1 to 57.17%, alongside substantial losses in ranking precision (mAP).

Table 5.11: Multi-task Cattle Re-Identification retrieval under SideView Protocol A on 69 unseen cattle identities evaluated against the 36,811-image fixed parlor gallery. Evaluated with released SideView target masks matching the oracle single-task reference (Run 6).^[E29,E36,E37,E38]

Configuration / Model	Rank-1 $\uparrow$	Rank-5 $\uparrow$	Rank-10 $\uparrow$	mAP $\uparrow$
<i>Barn-to-Parlor Video Queries (25,260 queries)</i>				
Single-Task Reference (Run 6)	63.90%	77.10%	82.58%	40.68%
Monolithic Hard-Shared (E1)	57.38%	73.33%	79.79%	30.37%
Modular Task-Private (E3)	49.08%	67.72%	75.03%	28.04%
PCGrad Hard-Shared (E4)	54.97%	68.81%	74.99%	30.23%
<i>Snapshots-to-Parlor Handheld Queries (607 queries)</i>				
Single-Task Reference (Run 6)	62.93%	75.29%	81.05%	40.42%
Monolithic Hard-Shared (E1)	57.17%	75.45%	82.70%	33.69%
Modular Task-Private (E3)	53.71%	72.32%	80.23%	28.78%
PCGrad Hard-Shared (E4)	57.00%	72.16%	78.25%	33.87%

Different results were obtained from the architecture and optimization approaches in Re-ID. The modular adapters (E3) approach decreased the performance to Barn Rank-1 of 49.08%. PCGrad (E4) improved the performance in comparison with the previous one, and achieved a result of Barn Rank-1 of 54.97%, and a higher value of Snapshot mAP of 33.87%. Nevertheless, its results were inconsistent with E1 and were still lower than those of the single-task network. Minor improvements

occurred at higher ranks, for example, Snapshot Rank-10 was slightly increased in E3. It can be concluded that there is a difference in the retrieval performance by the implementation of multiple tasks for feature sharing; however, this method is still not as accurate as a specific single-task Re-ID approach.

5.2.4 Observed Shared-Backbone Gradient Conflict

Table 5.12 reports the gradient conflict diagnostics recorded in E4 PCGrad training across all the 30 epochs, which consist of 16,140 super-steps. Overall, 44,177 pairwise conflict projections have been noted during training. The average number of conflict projections per super-step was 2.737, while the highest possible number was 6. There were also similarities among the conflict rates of the different task pairs. Conflicts between BCS and Behavior occurred in 47.8% of super-steps, BCS and Re-ID in 46.5%, and Behavior and Re-ID in 47.4%.

Table 5.12: Observed shared-backbone gradient conflict diagnostics recorded during E4 PCGrad multi-task training across all 30 epochs (16,140 super-steps). Conflicts were evaluated pairwise on the shared four-channel ResNet-18 parameter gradients prior to projection surgery.^[E35]

Diagnostic Metric / Evaluated Task Pair	Observed Training Quantity
BCS vs. Behavior Gradient Conflict Frequency	47.8% of super-steps
BCS vs. Re-ID Gradient Conflict Frequency	46.5% of super-steps
Behavior vs. Re-ID Gradient Conflict Frequency	47.4% of super-steps
Total Executed Multi-Task Super-Steps	16,140
Total Triggered Pairwise Conflict Projections	44,177
Mean Conflict Projections per Super-Step (out of 6 max)	2.737

As per the training diagnostics, conflicting gradients are observed quite often in the four-channel ResNet-18 backbone that is shared by both tasks. Nevertheless, one has to make sure that an essential distinction between E1 and E4 is not forgotten. E1 just summed up the gradients, whereas E4 applied the PCGrad method in order to determine the conflicting gradient directions and to project them. As a result, the E4 results demonstrate that conflicts are often observed only in the case of E4 training. They cannot be used in arguing that E1 experienced an equivalent rate of conflict or that gradient interference is the only explanation for the performance difference.

5.3 Analysis of Multi-Task Design Solutions

The empirical evaluations across single-task and multi-task configurations support distinct findings regarding the three multi-task design hypotheses:

1. **Monolithic Hard Parameter Sharing (E1):** For E1, all three tasks share the same ResNet-18 spatial backbone. In doing so, worse results were obtained for BCS, Behavior, and Re-ID. These findings demonstrate negative transfer since the performance of all three tasks does not necessarily improve when they

are learned together. It might be due to the fact that the information used by these three tasks is different: BCS relies on the body structure, Behavior on the movement and Re-ID on minute visual details. On the other hand, E1 cannot identify exactly which shared features were impacted and how they were impacted.

2. **Modular Task-Private Adapters (E3):** Adapter blocks were incorporated into E3 for each task between the shared backbone and the prediction head. Nonetheless, they did not solve consistently the problem of negative transfer. Some surveillance-related metrics and losses became better, while BCS and Re-ID performances declined, *along with Behavior balanced accuracy*. In addition to that, E3 contributed 395,904 parameters for training, thus enlarging the model capacity by 3.32%. As a result, any enhancement in performance cannot be attributed solely to task-specific routing.
3. **PCGrad Optimization Control (E4):** E4 modified the optimization strategy while maintaining the same model capacity. It had the same amount of trainable parameters as E1 - 11,926,706 - and didn't incorporate any adapter layers. PCGrad contributed to a reduction of negative transfer in several areas, such as Behavior accuracy, balanced accuracy, Macro-F1 score, and *CVB surveillance performance*. Moreover, it outperformed E3 in terms of important BCS and Re-ID metrics. Nevertheless, E4 performed worse than single-task models in some significant metrics.

Table 5.13 synthesizes the key design decisions and empirical dispositions established across the single-task and multi-task investigations.

The findings indicate that neither simple task-specific adapters nor gradient projection alone was sufficient to alleviate negative transfer effects in all three of the cattle monitoring tasks. The individual models were better in several task-specific measures. This corroborates the primary conclusion of the thesis that different tasks require different kinds of representations, and their combination in multi-task models may yield *different trade-offs depending on the metric*.

5.4 Statistical Analysis and Uncertainty Boundaries

All single-task and multi-task models in this chapter were evaluated based on only one trained checkpoint, and the reported results are point estimates. Evaluation with repeated training using different seeds, confidence intervals, and hypothesis testing was not done. Thus, the differences described above are purely descriptive comparisons that cannot be deemed statistically significant. The matched-sample design still yields fair internal comparisons since the same test data sets are used: 7,549 images of ScienceDB for BCS, 780 sequences for Behavior, and SideView Protocol A for Re-ID.

Nevertheless, such a comparison does not take into account variations of the different training seeds or the entire population variation. Furthermore, related video frames or image bursts are not supposed to be regarded as separate biological samples. Similarly, the PCGrad conflict rates provided in the Table 5.12 represent only one observation during the training process. Minor differences between metrics, like the

Table 5.13: Key design decisions established across single-task investigations and executed multi-task configurations.^[E01,E04,E06,E11,E13,E24,E27,E29,E31,E32,E33,E34,E35,E36,E37,E38]

Question / Domain	Empirical Evidence	Architecture / Method Disposition
ScienceDB grouping	Identity semantics and overlap audit	Repaired burst groups; sequence-safe evaluation.
Behavior source	Dense-video/provenance audit	CVB + Beef primary; Walking CVB-only.
Re-ID source	MultiCam access blocker	Approved SideView Protocol A benchmark.
BCS representation	Matched Run 1/4 comparison	RT-DETR crop + binary mask channel retained.
Behavior representation	Matched Run 2/5 comparison	Eight-frame RGB+mask + lightweight Conv1D retained.
Re-ID representation	Protocol A Run 3/6 comparison	Oracle GT crop+mask used as upper-bound condition.
Pose feature branch	Unreliable rear-view and lying poses	Left out of the primary baseline setup.
Viewpoint prior	Separate held-out test; transfer unverified	Kept as an independent test tool without feature fusion.
MTL sharing strategy	Held-out E1, E3, E4 evaluations	Hard sharing dropped scores across tasks compared to single-task baselines; PCGrad helped recover behavior scores.

0.18 percentage point difference in snapshot mAP between E4 and E1, cannot be considered equivalent.

5.5 Cross-Task Relationships and Claim Boundaries

Each of the three tasks is measured with various metrics. Such metrics are of various types and cannot be compared directly because of that.

- Metrics for BCS include physical Mean Absolute Error measured in BCS units, which means that smaller values are better, Acc@1 in ± 0.25 BCS units, and the measure of ordinal accuracy.
- Metrics for Behavior include accuracy, balanced accuracy, Macro-F1 across five classes, and F1 score for each of the classes separately. The higher the metric, the better.
- Metrics for Re-ID include Rank-1, Rank-5, Rank-10, and mean Average Precision (mAP).

As these measures cannot be adequately aggregated to give a combined score, negative transfer must be assessed for each task individually using comparison with its corresponding single-task model.

The evaluation cohorts also define strict generalization boundaries:

- **ScienceDB:** ScienceDB evaluates novel burst groups formed from automatic passages but does not validate them as novel biological cows.
- **Behavior:** Behavior evaluates protected video and sessions but not cow-disjoint herds. The Walking task is evaluated only on CVB surveillance data.
- **SideView Protocol A:** In the SideView Protocol A, the test dataset comprises 69 novel biological cows, which were not seen during the representation learning phase. However, the Re-ID models use the released ground-truth masks. It implies that the test results were gathered using oracle segmentation.

5.6 Discussion of Findings

The empirical findings presented in this chapter synthesize into three overarching insights:

The external data sets listed in the broader research project plan, including Ruchay, Dryad, MmCows, CBVD-5, and BECA, have yet to be validated and will undergo stress testing in the future. It means that just mentioning these data sets is not enough to imply that the model has learned to apply the knowledge gained to other data sets or environments.

In single-task experiments, cattle-centered visual data were beneficial against regular RGB data. Nonetheless, the amount of benefit was varied depending on the

task. Localized cropping and binary mask guidance helped with BCS by improving the closeness of the estimated physical score. They also helped with Re-ID by improving open set retrieval with challenging handheld camera viewpoints. Regarding Behavior, temporal foreground modeling enhanced class-balanced discrimination and detection of minority classes while redistributing accuracy among source distributions (Table [tab:behaviorcomparison](#)-[tab:behaviorsourcecomparison](#)). This shows the benefit of extracting animal foregrounds, even while making clear that the use of foreground information is not an absolute cure-all for each sub-metric. Second, hard parameter sharing resulted in consistent negative transfer effects at the level of performance across all three modalities. With respect to the respective single-task baselines, shared backbone learning for the three tasks increased BCS physical error (summary table), reduced Behavior balanced accuracy and minority class detection of walking behavior (summary table), and decreased Re-ID retrieval accuracy (summary table). Such consistent negative transfer effects do not indicate automatic benefit from hard sharing in these different tasks. Third, neither increased model capacity nor gradient projection addressed this task interference effect. Despite the use of separate residual bottleneck adapters for each task in E3, while the results of Behavior surveillance got slightly better, those of BCS and Re-ID were even worse. On the other hand, the gradient projection method applied in E4 provided an even better, yet much more selective improvement. It yielded the best results for Behavior among the multi-task models and managed to recover the performance lost in BCS ordinal scoring and Re-ID retrieval, while using the exact same amount of parameters as the hard-shared baseline. Moreover, the results from training presented in the summary table reveal that there were conflicts between the gradients of the tasks rather often during their joint training on the shared backbone. Nevertheless, the separate single-task models performed even better than the multi-task models with regard to all main metrics of the three tasks. Thus, the results did not reveal the usefulness of simultaneous learning of all tasks. In addition, the results revealed that different tasks require different kinds of representation and optimization, which cannot be provided by a shared backbone. That is why multi-task agricultural monitoring requires thorough metric comparison.

Chapter 6

Conclusion

6.1 Summary of Findings

The main aim of this thesis was to determine the effectiveness of cattle visual representations, in addition to multi-task learning, for monitoring dairy cattle. The tasks carried out under the thesis were: body condition scoring(BCS), behaviour recognition and re-identification of the cattle. The entire procedure took place through the process of record keeping, leakage-aware evaluation protocols, and matched-population controls to make sure its reproducible and reliable.

In the single-task experiments, it was possible to identify obvious advantages of cattle-oriented representations, but the degree of improvement was not equal across tasks and metrics.:

- As for BCS, there were 7,549 images in ScienceDB test set as detection and segmentation operated well on them, which constituted 93.89% of the whole test set. Cattle image cropping and binary masks yielded better performance in comparison with RGB representation. The value of MAE was reduced from 0.1929 to 0.1709 BCS units; Acc@1 within ± 0.25 was increased from 84.95% to 89.40%, however, exact classification was 43.57%. At the same time, balanced accuracy and Macro-F1 decreased slightly, from 40.23% to 39.70% and from 0.4074 to 0.4039, correspondingly.
- As for Behavior task, tests were conducted using 780 video clips consisting of 8 frames per clip that constitute 96.42% of matching data. Cropped cattle images, mask in the foreground, and small 1D temporal network were found to be beneficial in several ways compared to the single frame RGB model. Balanced accuracy increased to 74.43% from 71.30%, test loss decreased to 0.4430 from 0.5505, and walking F1 increased to 0.2456 from 0.2174. Nevertheless, the overall accuracy reduced to 87.44% from 88.46%. The approach was more effective for the feedlot camera images but somewhat less effective for the CVB surveillance images.
- In the case of Cattle Re-Identification, the model was evaluated on 69 unobserved cattle using the SideView Protocol A. Utilization of the ground-truth masks for creating target-centered crops enhanced first-match accuracy for both queries. In the case of Barn-to-Parlor, Rank-1 increased from 58.64% to 63.90%, while mAP increased from 38.32% to 40.68%. In the case of Snapshot-to-Parlor, Rank-1 increased from 38.88% to 62.93%, while mAP increased from

27.05% to 40.42%. The substantial gain in the snapshot performance indicates that the method can be very useful in cases where the animals’ poses and viewpoints significantly vary and illumination conditions are highly different. However, this assessment is oracle-based since ground truth masks were employed.

In the multi-task framework, training of three tasks together proved to be highly conflicting.:

- **Monolithic Hard-Shared Multi-Task Control (E1):** The Monolithic Hard-Shared Multi-Task Control (E1) enforced all three tasks to share the same four-channel ResNet-18 spatial backbone. The performance of all three tasks was degraded compared to the single-task models. For BCS, MAE rose from 0.1709 to 0.1788. For Behavior, balanced accuracy was reduced from 74.43% to 67.30% and Walking F1 was down from 0.2456 to 0.0408. For Re-ID, Barn Rank-1 was reduced from 63.90% to 57.38% and mAP fell from 40.68% to 30.37%.
- **Modular Task-Private Multi-Task Model (E3):** Modular Task-Private Multi-Task Learning (E3) added residual bottleneck adapters for each task between the shared backbone and task heads. It introduced 395,904 additional trainable parameters into the model, which is 3.32% higher compared to the capacity of E1. However, it was unable to avoid negative transfer on a general level. The loss of behavior test decreased to 0.4193 and accuracy of CVB surveillance increased to 80.33% in comparison to E1. At the same time, BCS became worse with MAE of 0.1916, and the Re-ID task performance became lower – 49.08% of Barn Rank-1 and 28.04% of mAP. The walking F1 score was equal to 0.0000 on the matched held-out test.
- **Gradient-Projected Hard-Shared Multi-Task Optimization Control (E4):** Gradient-Projected Hard-Shared Multi-Task Optimization Control (E4) was similar to E1 in number of 11,926,706 parameters. In addition, E4 applied PCGrad technique [23] only to the shared backbone during training. In case of E4 optimization, the number of projections with gradient conflicts is equal to 44,177 in 16,140 super-steps, wherein each super-step has 2.737 projections on average. It means that gradient conflicts happen quite often when training task pairs with the shared backbone in which conflict rates range from 46.5% to 47.8%. As for PCGrad comparison with E1, the model demonstrates accuracy equal to 86.92%, balanced accuracy equal to 69.15%, Macro-F1 equal to 0.7114, and walking F1 equal to 0.0909. Moreover, it managed to recover the important BCS and Re-ID results in contrast with E3. Still, E4 underperforms in comparison with the single-task models concerning some important metrics.

6.2 Answers to the Research Questions

The empirical findings from the single-task and multi-task investigations provide direct answers to the three research questions established in Chapter 1.

6.2.1 Research Question 1

How do cattle-centered representations affect BCS, behavior recognition, and Re-ID relative to their RGB single-task references?

As for the effect of cattle-centered representations compared to the RGB single-task models, the enhancement of the key prediction and retrieval results was achieved in all three tasks. Yet, the effect varies depending on the task and metric combination.:

- As for BCS, cattle-centered cropping and foreground mask help reduce the value of MAE by 0.0220 BCS units and improve Acc@1 by 4.45 percentage points in the matched test samples. The most significant improvements occurred in ordered error and tolerance metrics, while no improvements could be observed in the class-balanced metrics.
- As for Behavior Recognition, introducing the information about the foreground with temporal processing affected the results in terms of various classes and camera sources. The balanced accuracy has been improved on 3.13 percentage points, test loss decreased by 19.5%, and Walking F1 score has become 0.0282 higher. On the other hand, the overall accuracy became slightly lower by 1.02 percentage points, and the accuracy for CVB surveillance decreased as well. Thus, the use of the temporal foreground approach could be beneficial for some movement-related tasks, though there was no equal improvement in all the metrics.
- For Re-ID, the target-centered representation worked best under conditions when images had high variations. The Snapshot-to-Parlor Rank-1 has improved on 24.05 percentage points, and mAP metric on 13.37 percentage points. As for the barn surveillance case, Rank-1 has become better only on 5.26 percentage points.

Thus, the cattle-centered representations were useful, yet not equally effective for all the metrics, and they did not completely solve the problem of shortcut learning between the tasks.

The improvements mentioned above were made possible thanks to the application of more than one processing layer. Both Cropping and Masking layers were used in BCS and Re-ID, while Behavior utilized both of those in addition to Temporal Convolution layer. Thus, the improvement could not have been made thanks to segmentation alone.

6.2.2 Research Question 2

How can temporal information and task-specific visual cues be incorporated while preserving the different requirements of condition assessment, activity recognition, and identity matching?

This paper also analyzed how temporal features and visual cues can be integrated in consideration of diverse needs of condition scoring, behavior recognition and identity verification tasks.:

- For Body Condition Scoring, body shape is the key biological information. The use of bounding-box crops and binary foreground masks made it easier for the model to focus on the cow while retaining the biological information of

body shape that was essential for BCS. This improved prediction error without using information from multiple frames.

- In the case of Behavior Recognition, body pose and movement are both important aspects, thus requiring data from multiple frames for the model. In fact, a model using 8 frames with foreground guided TCN performed better than the single-frame model.
- For Cattle Re-ID system, details of the coat patterns and skin texture are required. The use of crops and segmentation masks played an important role in this regard. The Oracle masks turned out to be extremely useful in difficult conditions where the correct segmentation of the foreground was done.
- Anatomical pose keypoints and viewpoints have been thoroughly tested. The frozen SuperAnimal pose had a very poor detection rate (45.60% in behavior sequence cases and 72.92% in lying cattle cases) and was unable to generate the rear pelvic keypoints required for BCS. For these reasons, the pose information is not incorporated into the primary downstream pipeline. The real-cattle viewpoint had an 86.26% accuracy rate on the dataset created just for it, but it has not been incorporated into the primary models since it has not been tested in other farm environments.

Thus, visual information has to be chosen based on requirements for each specific task. The body shape and morphology are needed for BCS, for behavior recognition change over time is necessary, and unique coat surface details are necessary for re-identification. Thus, it would be more useful to do that instead of using the same visual information for all the tasks.

6.2.3 Research Question 3

How does task-conditioned sharing compare with hard sharing in task performance, negative transfer, and robustness under domain change?

Moreover, it was tested how well task-conditioned sharing and hard sharing affect the performance of tasks, negative transfer and robustness to domain shift. Comparing multi-task models with corresponding single-task models helped us make four important conclusions.:

- **Outcome-Level Negative Transfer under Hard Sharing:** Hard parameter sharing model (E1) performed worse than corresponding single-task models in all three tasks. In BCS, MAE rose by 0.0079. For Behavior, the balanced accuracy dropped by 7.13 percentage points while the F1 score for Walking fell to 0.0408. In Re-ID, Barn Rank-1 declined by 6.52 percentage points, while mAP decreased by 10.31 percentage points. From these findings, we note that there is task outcome negative transfer whereby different cattle vision tasks were restricted to use the same ResNet-18 spatial backbone.
- **Evaluation of Task-Conditioned Modularity:** The introduction of task-specific residual bottleneck adapters in E3 had little impact on addressing the negative transfer problem. When compared to E1, E3 helped in reducing test loss for Behavior from 0.6226 to 0.4193 and CVB surveillance accuracy increased from 76.78% to 80.33%. Nevertheless, BCS MAE rose to 0.1916, Barn

Rank-1 in Re-ID dropped to 49.08%, and Walking F1 decreased to 0.0000. In addition, E3 gained 395,904 trainable parameters, hence increasing its model capacity by 3.32% compared to E1. Due to this, the difference in performance cannot be justified purely based on the task-specific routing.

- **Evaluation of Gradient-Projected Optimization:** The evaluation of gradient-projected optimization was done using E4 where it applied PCGrad and had exactly the same number of parameters as E1 without adding additional capacity for the model. Conflicts between the gradients of task pairs occurred regularly throughout the process of training. In total, there were 44,177 conflict projections, out of 16,140 super-steps, where each super-step had an average of 2.737 projections and conflict rates ranging from 46.5% to 47.8%. PCGrad projected conflicting gradient components during optimization and provided selective performance mitigation. Behavior accuracy improved to 86.92%, an increase of 1.92 percentage points from E1. Balanced accuracy improved to 69.15%, an increase of 1.85 percentage points from E1, Macro-F1 improved to 0.7114 with an improvement of 0.0248, and Walking F1 improved to 0.0909 from 0.0408 in E1. It also recovered important BCS and Re-ID results compared to E3. In addition, E4 was unable to achieve full negative transfer avoidance. Single task models were superior in many ways. For BCS, MAE was 0.1709 compared to 0.1828 in E4. For Behavior, balanced accuracy was 74.43% compared to 69.15%, while walking F1 was 0.2456 compared to 0.0909. Finally, for Re-ID, Barn Rank-1 was 63.90% compared to 54.97%.
- **Robustness under Domain Change:** The multi-task models were evaluated under various circumstances, such as Barn and Snapshot views in Side-View and CVB surveillance compared to Kaggle Beef feedlot videos for Behavior. Nevertheless, these evaluations only occurred using the datasets utilized in this thesis and thus do not confirm any robustness to other external commercial herds.

6.3 Contributions of the Thesis

This thesis makes six contributions to cattle monitoring and multi-task representation learning.:

1. **Evaluation Protocols:** Several exact, reliable, and appropriate protocols have been developed to evaluate cattle monitoring tasks during the study process. The burst-group-disjoint protocol was adopted in the ScienceDB BCS case to prevent data leakage from the same passage frames. In Behavior Recognition using CVB and Kaggle Beef, data splitting was performed at both session and source levels. An identity-disjoint open-set protocol was employed in SideView Protocol A of Re-ID to validate 69 novel cows against a gallery of 36,811 parlor images.
2. **Single-Task Baselines on Standard Dataset:** Normal RGB inputs versus cattle-centered representation were compared in the study using the same test samples. This comparison highlighted the benefits that come with the use of

cattle cropping and foreground segmentation, providing substantial but task-dependent performance enhancement. For instance, MAE of BCS reduced from 0.1929 to 0.1709 and Snapshot Rank-1 of Re-ID increased from 38.88% to 62.93%.

3. **Task-Specific Cattle Input Representation:** The study suggested a task-specific approach whereby input representations should be designed based on task requirements. For BCS, static cow crops and masks were used for capturing body shape information. For Behavior, 1D temporal convolution was applied to an 8-frame cow video sequence. For Re-ID, essential surface patterns were retained in the images. The limitations in terms of pose estimation and viewpoint were also discussed.
4. **Benchmarking for Systematic Multi-Task Sharing:** In the systematic multi-task benchmark experiment, hard parameter sharing in E1, task-private residual adapter in E3, and gradient-projected optimization in E4 were compared to their respective single-task baselines. The same held-out test sets were used in the benchmark to evaluate the multi-task cattle vision system.
5. **Direct Gradient Conflict Diagnostics:** In this study, direct evidence of gradient conflicts in multi-task learning with PCGrad was presented. A total of 44,177 gradient conflicts were found in 16,140 super-steps. The rate of conflict is between 46.5% and 47.8%. This implies that conflicts occur regularly during multi-task cattle monitoring training.
6. **Multi-Task Trade-Offs:** As per the research findings, multi-task learning is not necessarily advantageous to all cattle monitoring tasks. In case tasks have shared features, one task will be improved while the other gets impaired. Negative transfer refers to the scenario where one solution improves one metric but not others.

6.4 Limitations

A number of limitations in the study need to be explicitly stated.:

1. **Absence of Verified Cow IDs in ScienceDB:** ScienceDB does not verify the IDs of cows. Though, the burst group disjoint split does not allow very similar frames from the same passage to appear in different sets, it does not guarantee that different sets have different cows.
2. **Session-Disjoint Rather than Cow-Disjoint Behavior Evaluation:** For Behavior task, the CVB and Kaggle Beef data have been split by videos and sessions rather than cows. Hence, Behavior task evaluation is session disjoint and not cow disjoint.
3. **Confounding of Walking class:** The Walking class was observed solely in the CVB surveillance dataset, where there were 26 test sequences. Therefore, there is some association between the camera source and the Walking action that could affect the classification performance.

4. **Automatic Perception Pipeline Coverage:** The coverage of the BCS perception pipeline was 93.89% with respect to the ScienceDB test set. In 8,040 test images, there were 489 detection failures and 2 segmentation failures. The Behavior pipeline retained 96.42% of the test sequences. Images with perception failure were excluded from our evaluation.
5. **Guidance from Oracle Segmentation in Re-ID:** In Run 6 and other multi-task Re-ID experiments, we relied on the oracle masks available in the dataset rather than on the output of a segmentation model. This provides an upper bound of the foreground isolation result, however, it does not reflect an automatic end-to-end solution.
6. **Joint Component Attribution:** The single-task models in Runs 4–6 were composed using different components like localization, cropping, binary mask, and temporal convolutions altogether. Hence, the contribution of each component was not measured individually.
7. **Confounding in Capacity of Modular Architecture:** E3 introduced 395,904 additional trainable parameters, increasing the model size by 3.32% relative to E1. For this reason, the performance difference between E1 and E3 is not only due to routing but because the architecture was also bigger.
8. **Boundary on Causal Explanation for Gradient Conflict:** Gradient conflict has been directly observed only when training E4 using PCGrad. The experiment shows the presence of gradient conflicts during E4 training. However, this is not sufficient to assert gradient conflict being the sole cause of worse performance in E1 as well as E1 having similar gradient conflict rate as E4.
9. **Point-Estimate Evaluation:** All single-task and multi-task models have been evaluated using only one checkpoint from training. Since computational limitations restricted our ability to run the training multiple times with different seeds, confidence intervals and hypotheses testing could not be conducted. Hence, all of the observed differences between the metrics are merely descriptive point comparisons.
10. **Untested Experiments in External Stress Testing Domain:** While we conducted an evaluation on the benchmark datasets, an external stress test on other types of datasets such as Ruchay, Dryad, MmCows, CBVD-5, and BECA datasets had been planned but was not executed. Hence, the obtained results cannot be considered for any verification of the cross-domain generalization.
11. **Unverified Downstream Transfer of Pose and Viewpoint:** While conducting the feasibility tests, both pose estimation and viewpoint were highly prone to failures as well as learning some source-specific shortcuts. Hence, the usefulness of them for downstream task datasets was not validated.
12. **Negative Transfer Persisting after Improvement of Results with PC-Grad:** Although some improvement was achieved with PCGrad, the performance of multi-task models remained inferior compared to the single-task models on many key metrics, showing that joint multi-task deployment requires an engineering trade-off.

6.5 Future Work

There are several potential research directions that could be pursued in the future in light of this paper’s results and limitations.:

- **Multiple Seed Training and Uncertainty Estimates:** In future research, it would be useful to repeat the training procedure with different seeds in order to obtain estimates for confidence intervals, variance and statistical testing of both one and multi-task models.
- **End-to-end Auto-Re-ID Segmentation:** In further research, an automatic segmentation model, e.g., fine-tuned SAM or YOLO-seg, could be used to generate prediction masks instead of oracle masks. It might help us to investigate how the errors made by the segmentation network affect the Re-ID task in realistic settings.
- **Generalization via External Testing with Multiple Herds:** In addition, the trained models should also be tested against independent external datasets. In case of BCS, it could be Ruchay2026 and Dryad, whereas in case of Behavior Recognition, MmCows and CBVD-5 would do.
- **Cow Identity Retrieval over Time:** One possible approach for doing future work may be through cow Re-ID over time via the datasets BECA-L and BECA-D. It would be able to reveal the stability of cow identity attributes across different groupings, lactations, season coat, and body development phases.
- **Ablation Study of Individual Perception Components:** An ablation study for the effect of bounding box, soft mask, binary foreground mask, and sequence lengths could be performed.
- **Robust Pose Estimation of Animal:** Fine-tuning of animal pose estimation models for the laying and rear view of the cattle could be carried out.
- **Multi-task Optimization Approaches:** One can perform future experiments using partial parameter sharing approaches like branched trunk and adaptive loss weighting approaches like GradNorm [10] to resolve the conflict of gradients between different tasks.

6.6 Conclusion

The thesis was successful in establishing an empirical and methodological framework for multi-task deep learning in the context of precision cattle monitoring, involving body condition scoring, behavior recognition, and cow re-identification under controlled single-task and joint multi-task settings.

Evidence from experiments has shown that cow-focused visual representations offer significant utility in the performance of specific individual livestock monitoring tasks by improving BCS error tolerance and improving the re-identification of cows when there are unconstrained shifts in the cameras. This is however task dependent and does not address the separate needs of representation for morphological regression, temporal modeling, and biometric identification.

Utilizing hard parameter sharing in the multi-task model resulted in negative transfer in all three tasks. The addition of task-private residual adapters did not solve the performance issue. According to PCGrad, gradients of different tasks on the shared backbone network usually have contradictory directions during training. Thus, by managing contradictory parts of these gradients, PCGrad could mitigate the issue to a certain degree. In particular, PCGrad contributed significantly to improving the performance of Behavior classification and CVB surveillance. Nevertheless, the single-task model surpassed the performance of the multi-task one according to some metrics specific for a particular task. Hence, the findings indicate that shared representation in the case of multi-task learning is not always advantageous for agricultural computer vision. A practical multi-task system requires a good balance between the shared cattle representation and task-specific processing.

Bibliography

- [1] A. J. Edmonson, I. J. Lean, L. D. Weaver, T. Farver, and G. Webster, “A body condition scoring chart for holstein dairy cows,” *Journal of Dairy Science*, vol. 72, no. 1, pp. 68–78, 1989. DOI: 10.3168/jds.S0022-0302(89)79081-0
- [2] J. D. Ferguson, D. T. Galligan, and N. Thomsen, “Principal descriptors of body condition score in Holstein cows,” *Journal of Dairy Science*, vol. 77, no. 9, pp. 2695–2703, 1994. DOI: 10.3168/jds.S0022-0302(94)77212-X
- [3] M. A. G. von Keyserlingk, J. Rushen, A. M. de Passillé, and D. M. Weary, “Invited review: The welfare of dairy cattle—key concepts and the role of science,” *Journal of Dairy Science*, vol. 92, no. 9, pp. 4101–4111, 2009. DOI: 10.3168/jds.2009-2326
- [4] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778. DOI: 10.1109/CVPR.2016.90
- [5] I. Misra, A. Shrivastava, A. Gupta, and M. Hebert, “Cross-stitch networks for multi-task learning,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 3994–4003. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2016/html/Misra_Cross-Stitch_Networks_for_CVPR_2016_paper.html
- [6] W. Andrew, S. L. Hannuna, N. W. Campbell, and T. Burghardt, “Automatic individual holstein friesian cattle identification via selective local coat pattern matching in RGB-D imagery,” in *2016 IEEE International Conference on Image Processing (ICIP)*, Conference held in 2016; institutional record lists publication in 2017, 2017, pp. 484–488. DOI: 10.1109/ICIP.2016.7532404
- [7] K. He, G. Gkioxari, P. Dollár, and R. Girshick, “Mask R-CNN,” in *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 2961–2969. [Online]. Available: https://openaccess.thecvf.com/content_iccv_2017/html/He_Mask_R-CNN_ICCV_2017_paper.html
- [8] S. Bai, J. Z. Kolter, and V. Koltun, *An empirical evaluation of generic convolutional and recurrent networks for sequence modeling*, 2018. arXiv: 1803.01271. [Online]. Available: <https://arxiv.org/abs/1803.01271>
- [9] S. Beery, G. Van Horn, and P. Perona, “Recognition in terra incognita,” in *Proceedings of the European Conference on Computer Vision*, 2018. [Online]. Available: https://openaccess.thecvf.com/content_ECCV_2018/html/Beery_Recognition_in_Terra_ECCV_2018_paper.html

- [10] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich, “GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks,” in *Proceedings of the 35th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 80, 2018, pp. 794–803. [Online]. Available: <https://proceedings.mlr.press/v80/chen18a.html>
- [11] A. Kendall, Y. Gal, and R. Cipolla, “Multi-task learning using uncertainty to weigh losses for scene geometry and semantics,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7482–7491. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2018/html/Kendall_Multi-Task_Learning_Using_CVPR_2018_paper.html
- [12] A. Mathis et al., “DeepLabCut: Markerless pose estimation of user-defined body parts with deep learning,” *Nature Neuroscience*, vol. 21, pp. 1281–1289, 2018. DOI: 10.1038/s41593-018-0209-y
- [13] J. Rodríguez Álvarez et al., “Body condition estimation on cows from depth images using convolutional neural networks,” *Computers and Electronics in Agriculture*, vol. 155, pp. 12–22, 2018. DOI: 10.1016/j.compag.2018.09.039
- [14] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, “CBAM: Convolutional block attention module,” in *Proceedings of the European Conference on Computer Vision*, 2018, pp. 3–19. [Online]. Available: <https://arxiv.org/abs/1807.06521>
- [15] S. Yan, Y. Xiong, and D. Lin, “Spatial temporal graph convolutional networks for skeleton-based action recognition,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, 2018. DOI: 10.1609/aaai.v32i1.12328
- [16] C. Feichtenhofer, H. Fan, J. Malik, and K. He, “SlowFast networks for video recognition,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 6202–6211. [Online]. Available: https://openaccess.thecvf.com/content_ICCV_2019/html/Feichtenhofer_SlowFast_Networks_for_Video_Recognition_ICCV_2019_paper.html
- [17] S. Liu, E. Johns, and A. J. Davison, “End-to-end multi-task learning with attention,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 1871–1880. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2019/html/Liu_End-To-End_Multi-Task_Learning_With_Attention_CVPR_2019_paper.html
- [18] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 97, 2019, pp. 6105–6114. [Online]. Available: <https://proceedings.mlr.press/v97/tan19a.html>
- [19] W. Cao, V. Mirjalili, and S. Raschka, “Rank consistent ordinal regression for neural networks with application to age estimation,” *Pattern Recognition Letters*, vol. 140, pp. 325–331, 2020. DOI: 10.1016/j.patrec.2020.11.008
- [20] M. Crawshaw, *Multi-task learning with deep neural networks: A survey*, 2020. arXiv: 2009.09796. [Online]. Available: <https://arxiv.org/abs/2009.09796>
- [21] R. Geirhos et al., “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020. DOI: 10.1038/s42256-020-00257-z

- [22] T. Standley, A. Zamir, D. Chen, L. Guibas, J. Malik, and S. Savarese, “Which tasks should be learned together in multi-task learning?” In *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119, 2020, pp. 9120–9132. [Online]. Available: <https://proceedings.mlr.press/v119/standley20a.html>
- [23] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn, “Gradient surgery for multi-task learning,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020. [Online]. Available: <https://proceedings.neurips.cc/paper/2020/hash/3fe78a8acf5fda99de95303940a2420c-Abstract.html>
- [24] W. Andrew, J. Gao, S. Mullan, N. Campbell, A. W. Dowsey, and T. Burghardt, “Visual identification of individual Holstein-Friesian cattle via deep metric learning,” *Computers and Electronics in Agriculture*, vol. 185, 2021, Art. no. 106133. DOI: 10.1016/j.compag.2021.106133
- [25] J. Gao, T. Burghardt, W. Andrew, A. W. Dowsey, and N. W. Campbell, *Towards self-supervision for video identification of individual Holstein-Friesian cattle: The Cows2021 dataset*, 2021. arXiv: 2105.01938. [Online]. Available: <https://arxiv.org/abs/2105.01938>
- [26] P. W. Koh, S. Sagawa, H. Marklund, et al., “WILDS: A benchmark of in-the-wild distribution shifts,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 5637–5664. [Online]. Available: <https://proceedings.mlr.press/v139/koh21a.html>
- [27] K. Xiao, L. Engstrom, A. Ilyas, and A. Madry, “Noise or signal: The role of image backgrounds in object recognition,” in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://arxiv.org/abs/2006.09994>
- [28] Z. Weng, F. Meng, S. Liu, Y. Zhang, Z. Zheng, and C. Gong, “Cattle face recognition based on a two-branch convolutional neural network,” *Computers and Electronics in Agriculture*, vol. 196, 2022, Art. no. 106871. DOI: 10.1016/j.compag.2022.106871
- [29] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns*, vol. 4, no. 9, 2023, Art. no. 100804. DOI: 10.1016/j.patter.2023.100804
- [30] A. Kirillov et al., “Segment anything,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4015–4026. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2023/html/Kirillov_SegmentAnything_ICCV_2023_paper.html
- [31] K. E. Ong, S. Retta, R. Srinivasan, S. Tan, and J. Liu, “CattleEyeView: A multi-task top-down view cattle dataset for smarter precision livestock farming,” in *2023 IEEE Visual Communications and Image Processing*, 2023, pp. 1–5. DOI: 10.1109/VCIP59821.2023.10402676 [Online]. Available: <https://arxiv.org/abs/2312.08764>
- [32] A. Zia et al., *CVB: A video dataset of cattle visual behaviors*, 2023. arXiv: 2305.16555. [Online]. Available: <https://arxiv.org/abs/2305.16555>

- [33] K. Li, D. Fan, H. Wu, and A. Zhao, “A new dataset for video-based cow behavior recognition,” *Scientific Reports*, vol. 14, 2024, Art. no. 18702. DOI: 10.1038/s41598-024-65953-x
- [34] N. Ravi et al., *SAM 2: Segment anything in images and videos*, 2024. arXiv: 2408.00714. [Online]. Available: <https://arxiv.org/abs/2408.00714>
- [35] H. Vu et al., “MmCows: A multimodal dataset for dairy cattle monitoring,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024. [Online]. Available: https://papers.nips.cc/paper_files/paper/2024/hash/6d8f3f71b22f9d2e9320d7bdb73acea7-Abstract-Datasets_and_Benchmarks_Track.html
- [36] S. Ye et al., “SuperAnimal pretrained pose estimation models for behavioral analysis,” *Nature Communications*, vol. 15, 2024, Art. no. 5165. DOI: 10.1038/s41467-024-48792-2
- [37] Y. Zhao et al., “DETRs beat YOLOs on real-time object detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16 965–16 974. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Zhao_DETRs_Beat_YOLOs_on_Real-time_Object_Detection_CVPR_2024_paper.html
- [38] V. Antognoli, L. Presutti, M. Bovo, D. Torreggiani, and P. Tassinari, “Computer vision in dairy farm management: A literature review of current applications and future perspectives,” *Animals*, vol. 15, no. 17, 2025, Art. no. 2508. DOI: 10.3390/ani15172508
- [39] I. Attri et al., “SAAM-VetNet: An attention-based multi-task framework for animal disease detection and severity grading,” *Annals of Medicine and Surgery*, vol. 87, no. 10, pp. 6399–6408, 2025. DOI: 10.1097/MS9.00000000000003728
- [40] X. Feng, J. Zhang, Y. Qi, L. Liu, and Y. Li, “CATR-Net: Cattle-attentive transformer with adaptive and enhanced segmentation and recognition,” *Computers and Electronics in Agriculture*, vol. 239, 2025, Art. no. 111038. DOI: 10.1016/j.compag.2025.111038
- [41] F. Liu, Y. Zhang, Y. Liu, J. Li, M. Li, and J. Yao, “A novel lightweight dairy cattle body condition scoring model for edge devices based on tail features and attention mechanisms,” *Veterinary Sciences*, vol. 12, no. 9, 2025, Art. no. 906. DOI: 10.3390/vetsci12090906
- [42] O. Palma, L. M. Plà-Aragónés, A. Mac Cawley, and V. M. Albornoz, “Ai and data analytics in the dairy farms: A scoping review,” *Animals*, vol. 15, no. 9, 2025, Art. no. 1291. DOI: 10.3390/ani15091291
- [43] P. Yu, T. Burghardt, A. W. Dowsey, and N. W. Campbell, “Holstein-Friesian re-identification using multiple cameras and self-supervision on a working farm,” *Computers and Electronics in Agriculture*, vol. 237, 2025, Art. no. 110568. DOI: 10.1016/j.compag.2025.110568 [Online]. Available: <https://arxiv.org/abs/2410.12695>
- [44] M. Asim, B. Anam, M. N. Ali, and B. S. Kim, “Cattle farming activity monitoring using advanced deep learning approach,” *Sensors*, vol. 26, no. 3, 2026, Art. no. 785. DOI: 10.3390/s26030785

- [45] W. Grolleau, A. Chaouch, A. Sabourin, G. Lapouge, and C. Achard, *MOO: A multi-view oriented observations dataset for viewpoint analysis in cattle re-identification*, 2026. arXiv: 2603.04314. [Online]. Available: <https://arxiv.org/abs/2603.04314>
- [46] O. Guzhva, E. Ternman, M. Lindberg, E. Telezhenko, and C. Kronqvist, “Pick-AMoo: LIDAR-enhanced mask R-CNN segmentation for precision weight estimation in dairy cattle using smartphone imaging,” *Scientific Reports*, vol. 16, 2026, Art. no. 16020. DOI: 10.1038/s41598-026-54742-3
- [47] M. Lee, L. O. Tedeschi, and S. Seo, “Biosensors in precision livestock farming in dairy production: Decoding animals’ needs,” *Animal Bioscience*, vol. 39, no. 4, 2026, Art. no. 260154. DOI: 10.5713/ab.260154
- [48] M. I. Sani, M. van der Voort, B. Tekinerdogan, and H. Hogeveen, “Application of precision livestock farming: Challenges and opportunities,” *Animal Bioscience*, vol. 39, no. 4, 2026, Art. no. 250895. DOI: 10.5713/ab.250895
- [49] L. Yao et al., “Automated dairy cattle body condition score using side-view images and deep learning,” *Journal of Dairy Science*, vol. 109, no. 9, pp. 9446–9458, 2026. DOI: 10.3168/jds.2025-27759
- [50] Y. Zhang et al., “BECA: A computer vision dataset for long-term recognition in beef cattle,” *Scientific Data*, vol. 13, 2026, Art. no. 7. DOI: 10.1038/s41597-025-06326-5
- [51] T. T. Zin and P. Tin, “Computer vision in precision livestock farming: Artificial intelligence-driven technologies and applications for sustainable animal production,” *Animal Bioscience*, vol. 39, no. 4, 2026, Art. no. 260165. DOI: 10.5713/ab.260165

Appendix A

Dataset and Experiment Evidence

A.1 Versioned Evidence Register

The document was checked against Git commit `d55ea2df436f504c3de4d83bcf9db845a6b706e3`. It is the latest Git commit examined while writing this document, although individual experiment artifacts may come from an earlier or unknown execution revision. None of the models was rerun during the thesis rewrite.

ID	Source	Use in the manuscript
E01	<code>phase3_canonical_roadmap.md</code>	Scope, scientific direction, and dataset roles.
E02	<code>memory/state.md</code>	Latest project and experiment status.
E03	<code>phase3_deadline_execution_2026-09-26.md</code>	Focused eight-run plan and E1/E3 roles.
E04	<code>datasets/bcs/sciencedb/split_report.md</code>	ScienceDB split counts and identity limitation.
E05	<code>datasets/behavior/cvb_beef/split_report.md</code>	Behavior mapping, grouping, and split counts.
E06	<code>datasets/id/sideviewcows2026/split_report.md</code>	SideView protocols and identity separation.
E07	<code>datasets/dataset_registry.csv</code>	Dataset roles, modalities, and availability.
E08	<code>artifacts/bcs_baseline/bcs_baseline_metrics.json</code>	Run 1 metrics and configuration.
E09	<code>artifacts/behavior_baseline/behavior_baseline_metrics.json</code>	Run 2 metrics.
E10	<code>scripts/train_sciencedb_bcs_baseline.py</code>	Ordinal-BCE implementation.
E11	<code>docs/audits/phase3_perception_feasibility.md</code>	Detector, mask, pose, and manual-review evidence.
E12	<code>scripts/train_cvb_beef_behavior_baseline.py</code>	Run 2 implementation.
E13	<code>docs/research_log/2026-09-23_behavior_primary_stack_correction.md</code>	Behavior dataset-role correction.
E14	<code>docs/research_log/2026-09-23_sciencedb_bcs_baseline_full_training_results.md</code>	Run 1 execution record.
E15	<code>artifacts/behavior_baseline/behavior_baseline_30epoch_summary.md</code>	Run 2 execution summary.
E16	<code>thesis_marking_rubrics.md</code>	BRACU report criteria.
E17	<code>thesistemplate/main.tex</code>	Supplied report structure.
E18	<code>cattle_thesis_p2_latex/core/titlepage.tex</code>	Draft author roster.

ID	Source	Use in the manuscript
E19	docs/research_log/2026-09-23_sideviewcows2026_reid_baseline_smoke_test.md	Run 3 smoke validation.
E20	scripts/train_sideview_reid_baseline.py	Run 3 classifier and retrieval implementation.
E21	artifacts/reid_baseline_smoke/reid_baseline_metrics.json	Run 3 smoke scope only.
E22	artifacts/reid_baseline/reid_baseline_metrics.json	Run 3 full Protocol A results.
E23	docs/research_log/2026-09-23_sideviewcows2026_reid_baseline_full_training_results.md	Run 3 full-run record.
E24	artifacts/bcs_perception_run4/bcs_perception_matched_test_comparison.json	Run 4 coverage and matched comparison.
E25	scripts/train_sciencedb_bcs_perception.py	Run 4 four-channel model.
E26	artifacts/behavior_perception_cache/perception_summary.json	Run 5 cache coverage and exclusions.
E27	artifacts/behavior_run5_test/run2_vs_run5_matched_comparison.json	Run 5 matched task, class, and source metrics.
E28	scripts/train_cvb_beef_behavior_tcn.py	Run 5 temporal model and sampling.
E29	artifacts/reid_perception_run6/reid_perception_metrics.json	Run 6 representation and Protocol A results.
E30	scripts/train_sideview_reid_perception.py	Run 6 oracle crop-and-mask implementation.
E31	artifacts/viewpoint_real_finetune/test_evaluation_metrics.json	Held-out real-cattle viewpoint result.
E32	docs/research_log/2026-09-24_sciencedb_bcs_perception_full_training_results.md	Run 4 execution narrative.
E33	docs/research_log/2026-09-24_run5_behavior_perception_test_evaluation_results.md	Run 5 strict test-gate narrative.
E34	docs/research_log/2026-09-24_sideviewcows2026_reid_perception_run6_full_training_results.md	Run 6 full-run narrative.
E35	artifacts/mtl_e4_training/mtl_e4_metrics.json	E4 training diagnostics; 16,140 super-steps; 44,177 projections.
E36	artifacts/mtl_e1_evaluation/mtl_e1_test_evaluation_metrics.json	Official E1 held-out BCS, Behavior, and Re-ID results.
E37	artifacts/mtl_e3_evaluation/mtl_e3_test_evaluation_metrics.json	Official E3 held-out BCS, Behavior, and Re-ID metrics.
E38	artifacts/mtl_e4_evaluation/mtl_e4_test_evaluation_metrics.json	Official E4 held-out BCS, Behavior, and Re-ID metrics.
E39	scripts/train_mtl_e1_hard_shared.py	Monolithic hard-shared architecture and training procedure.
E40	scripts/train_mtl_e3_modular.py	Task-private residual adapter architecture (+3.32% capacity).
E41	scripts/train_mtl_e4_pcgrad.py	PCGrad gradient projection surgery implementation.

A.2 Canonical Evaluation Boundaries

ScienceDB contains 37,045 training images, 8,481 validation images, and 8,040 test images, organized into 3,958, 850, and 845 repaired burst groups, respectively. This setup supports sequence-safe evaluation rather than a cow-disjoint evaluation. The Behavior protocol includes 3,785 training, 680 validation, and 809 test samples grouped according to 267 source videos or sessions. Walking is available only in the CVB dataset. SideView Protocol A is trained using 41 cows and evaluated on 69 different cows, with a 36,811-image parlor gallery, 25,260 barn queries, and 607 snapshot queries.^[E04,E05,E06]

Runs 4 and 5 employ matched controls because their perception processing streams exclude certain canonical examples. Run 4 and its Run 1 control have 7,549 test images in common. Run 5 and its Run 2 control have 780 test sequences in common. Run 6 includes all inputs from Protocol A because SideView provides paired ground-truth masks, but the result of Run 6 represents an oracle condition rather than automatic segmentation.^[E24,E27,E29]

A.3 Remaining Evidence Boundaries

The baselines and multi-task models were validated using just one trained checkpoint. Due to computational restrictions, the research could not perform repeated-seed analysis, calculate confidence intervals, or carry out hypothesis tests. The cross-dataset experiments on uncalibrated herds have not been performed yet and remain a task for the future. For the Re-ID experiments, the provided ground-truth masks were used instead of an automated upstream segmentation model. It has not yet been checked whether anatomical pose estimation and viewpoint priors can be transferred to other downstream tasks. Institutional approval is required for all administrative, committee, ethics, and AI-related disclosures.

A.4 Model Names and Repository Identifiers

Descriptive model names are used in the methodology. The identifiers listed below preserve the connection to the result tables and archived experiment records for all completed single-task and multi-task configurations.

Identifier	Model name
Run 1	Baseline RGB BCS model.
Run 2	Baseline single-frame RGB behavior model.
Run 3	Baseline RGB Re-ID model.
Run 4	Perception-enhanced BCS model.
Run 5	Perception-enhanced temporal behavior model.
Run 6	Oracle segmentation-guided Re-ID model.
Run 7 / E1	Monolithic hard-shared multi-task control.
Run 8 / E3	Modular task-private multi-task model.
E4	PCGrad hard-shared multi-task optimization control.

A.5 Methodology Evidence Cross-Reference

The following table categorizes the supporting records for Chapter 4. These evidence identifiers refer to the versioned register included in this appendix. This categorization replaces repeated record references in the methodology text without changing the underlying sources.

Methodology topic	Evidence identifiers
Baseline RGB Body Condition Scoring	E08, E10, E14.
Perception-Enhanced Body Condition Scoring	E24, E25.
Baseline Single-Frame RGB Behavior Recognition	E05, E09, E12, E15.
Perception-Enhanced Temporal Behavior Recognition	E26, E28, E27.
Baseline RGB Cattle Re-Identification	E06, E20, E22, E23.
Oracle Segmentation-Guided Re-Identification	E29, E30.
Monolithic Hard-Shared Multi-Task Control	E36, E39.
Modular Task-Private Multi-Task Model	E37, E40.
PCGrad Hard-Shared Optimization Control	E35, E38, E41.
Data Sources and Scientific Roles	E04, E05, E13, E06.
Perception Feasibility and Component Selection	E11, E31.
Experimental model summary	E02, E03, E22, E24, E27, E29, E35, E36, E37, E38.

Appendix B

Reproducibility and Evidence Boundaries

B.1 Reproducibility Boundary

In order to ensure reproducibility, the exact data splitting records, code used in the experiments, performance results in machine readable format and task-specific information is provided in the thesis itself. Larger resources like the original datasets, fully-trained checkpoints and cloud repository contents are not included in the thesis. It is thus clear that in order to repeat the experiments one needs access to the right version of the repository, manifest and dataset files.

Reproducibility details were stored separately for each run. For Run 1, seed 42 and split hashes, and for Run 5 - seed 2026 and retained data sample hashes were stored. Also, the checkpoint reload for Run 5 was checked to be bit-exact relative to the stored version. The same checkpoint check was performed for Run 6, along with the synchronization of RGB and mask crops. For the multi-task comparison, E1, E3, and E4 used fixed implementation versions, and checkpoint selection was based only on validation performance. They are evaluated on exactly the same held-out samples to allow direct comparison. E4 provides additional reproducibility details by storing random seeds of PCGrad as well as training diagnostics. In some old runs, it was not possible to save the original Git commit due to lack of Git metadata in the container. Instead of speculating or providing the missing details later, unknown values are kept in these fields, and thesis-review SHA is not used as a substitute.^[E08,E27,E29,E35,E36,E37,E38,E39,E40,E41]

B.2 Scientific Interpretation Boundaries

Independence units. ScienceDB keeps connected bursts separate across the evaluation splits rather than separating the data based on individual cow identities. CVB and Beef are organized based on source videos or sessions rather than verified individual cows. Only SideView Protocol A supports a comparison where the cow identities used for learning and evaluation are different.^[E04,E05,E06]

Matched coverage. Run 4 includes 7,549 of 8,040 test images, and Run 5 includes 780 of 809 test sequences. Their corresponding RGB checkpoints were evaluated on the same retained samples. The lack of full coverage is still a limitation of the

system.^[E24,E27]

Combined configurations. Each of runs 4, 5, and 6 alters multiple components simultaneously. For example, run 4 alters the localization, cropping, and mask components; run 5 alters the representation and temporal aggregation components; and run 6 alters the crop design and uses a ground truth mask channel. This makes it difficult to isolate the contribution of each individual component.

Oracle Re-ID condition. In Run 6, the ground-truth target masks are provided by SideView. The results do not demonstrate automatic SAM segmentation, nor do they prove that shortcut learning has been eliminated.^[E29,E30]

Capacity confounding in modular MTL. The modular MTL architecture (E3) introduced 395,904 additional parameters that increased the capacity of the model by 3.32% relative to E1. For that reason, performance gaps can be attributed not only to the novel architecture but also to the increase in size of the model.^[E37,E40]

Gradient conflict diagnostics. Gradient conflicts were measured only for the E4 PCGrad method; opposing gradients were observed between tasks, yet no statistics can now be applied to E1 or E3.^[E35,E41]

Selective mitigation. It was beneficial for tackling negative transfer for specific instances where behavior recognition was considered. The PCGrad approach was not effective for improving all metrics, and single-task models proved to be superior in many cases. Thus, negative transfer could not be fully addressed.^[E36,E38]

Statistical scope. This paper presents the results of a single experiment run for the single task and multitask models. The confidence intervals, standard deviation, p-value, and statistical significance of the findings are unknown due to the lack of repeated seed experiments. Generalization to novel data (herds) has not been confirmed at the present time.

B.3 Open Deliverables

The future experiments will be performed by validating the models on different sets of cows from various datasets other than those used in the training process. The datasets to be used in such validations will include Ruchay, Dryad, MmCows, CBVD-5, and BECA. In addition, the models will also be tested under different random seeds to measure the variability of their accuracies.

This verification process is mandatory in order to prove that the authorship order, semester, committee information, ethical and AI considerations, permissions for the datasets, acknowledgments, and all other IEEE format requirements have been fulfilled in accordance with the standards. If there are any signatures required, then those must be done.